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(54) **COMPOSITIONS AND METHODS FOR
TREATMENT OF LUNG FUNCTION**

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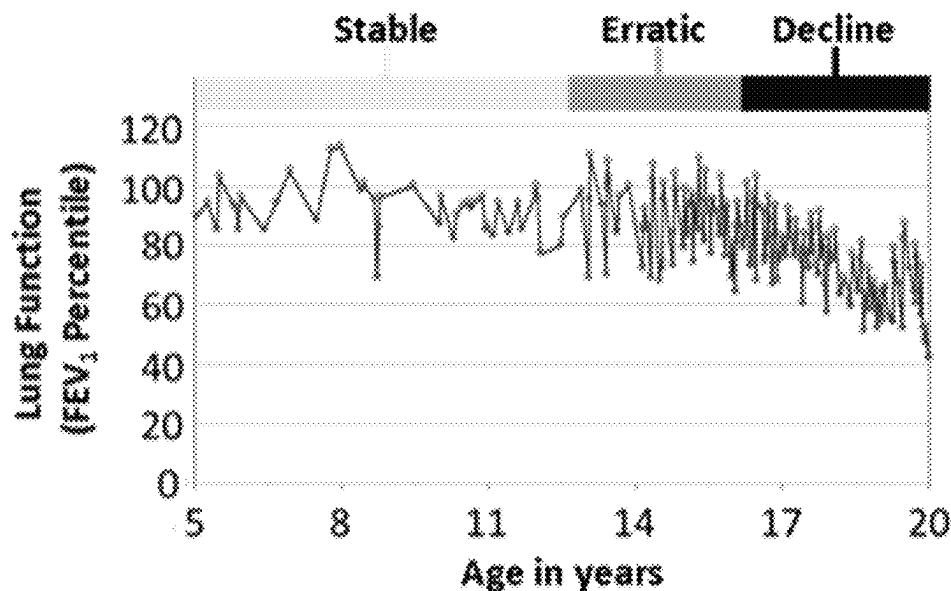
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(57) **ABSTRACT**

Disclosed herein are methods for detecting protein expres-
sion in an individual diagnosed with cystic fibrosis. The
methods, in certain aspects, include the steps of obtaining a
sample from said individual and detecting expression in said
sample of each protein of a protein set. The method may
further include the step of determining expression level of
one or more proteins of the protein set. The disclosed
methods may be used to predict one or more clinical
parameters in an individual having cystic fibrosis.

24 Claims, 16 Drawing Sheets



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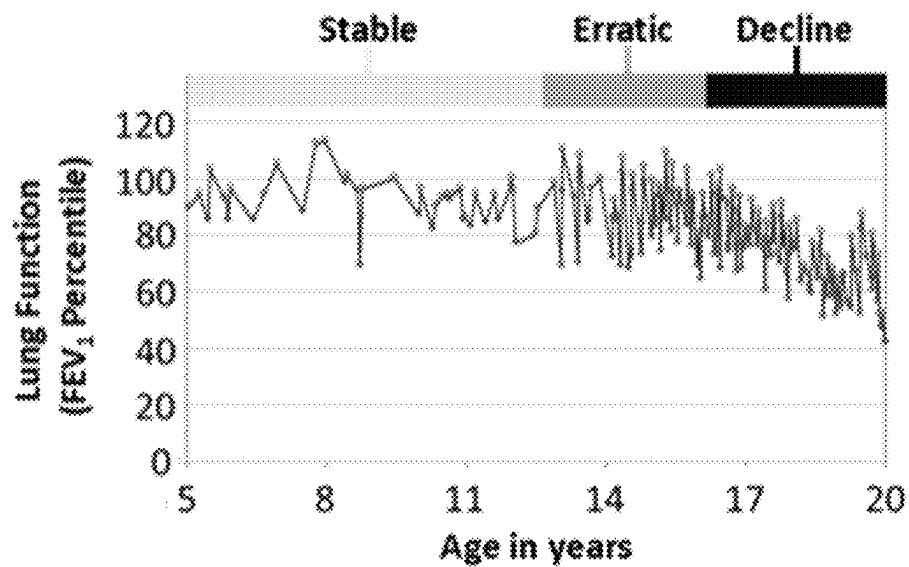
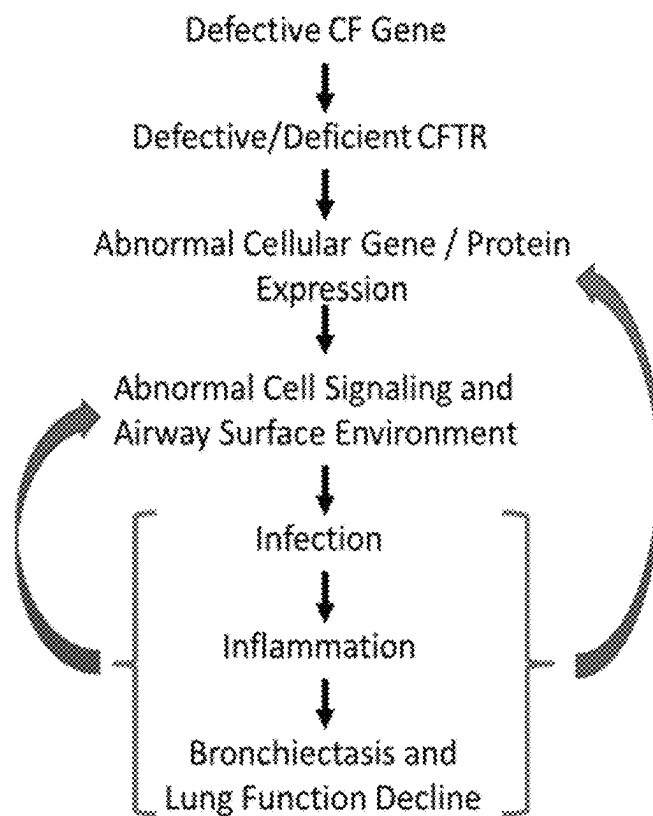
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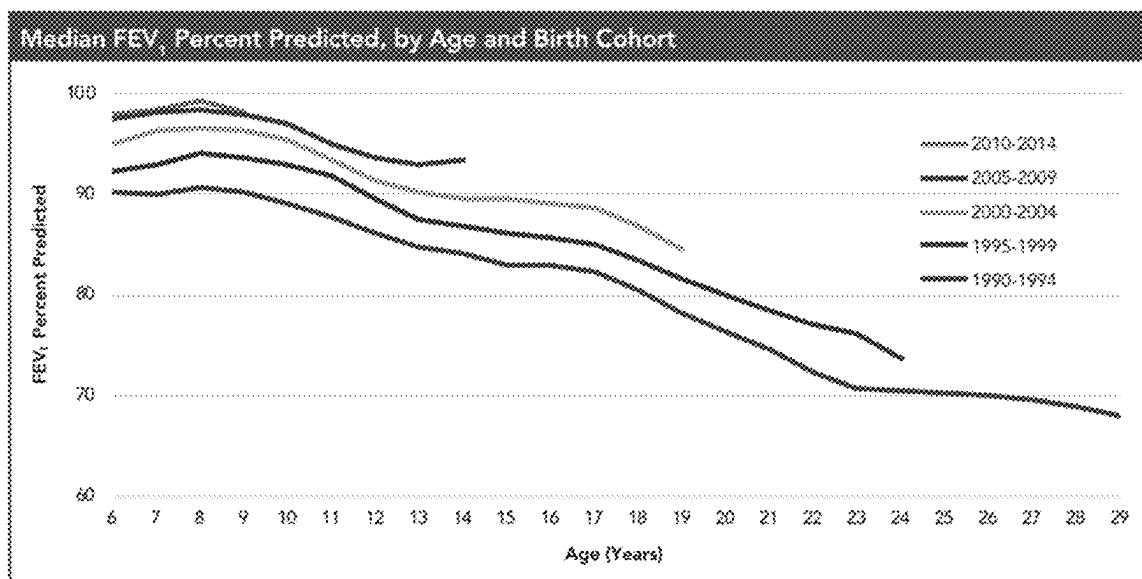
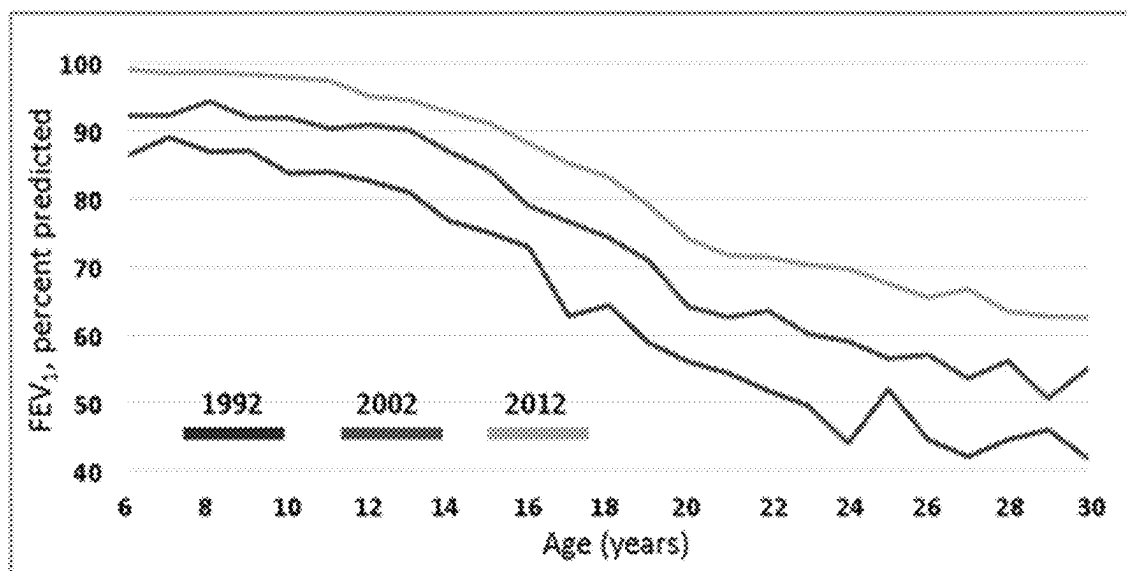
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**FIG. 1****FIG. 2**

**FIG. 3**

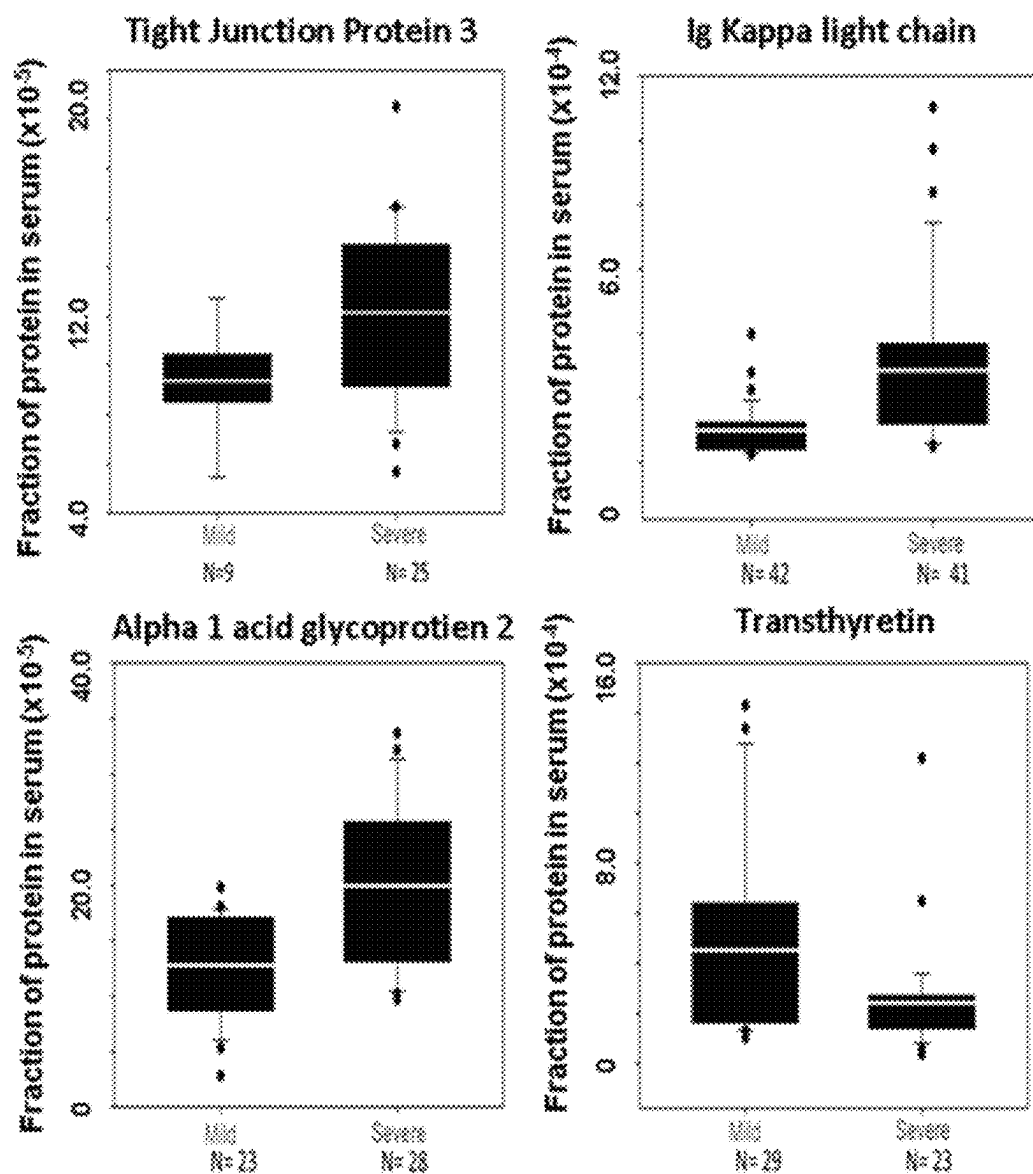
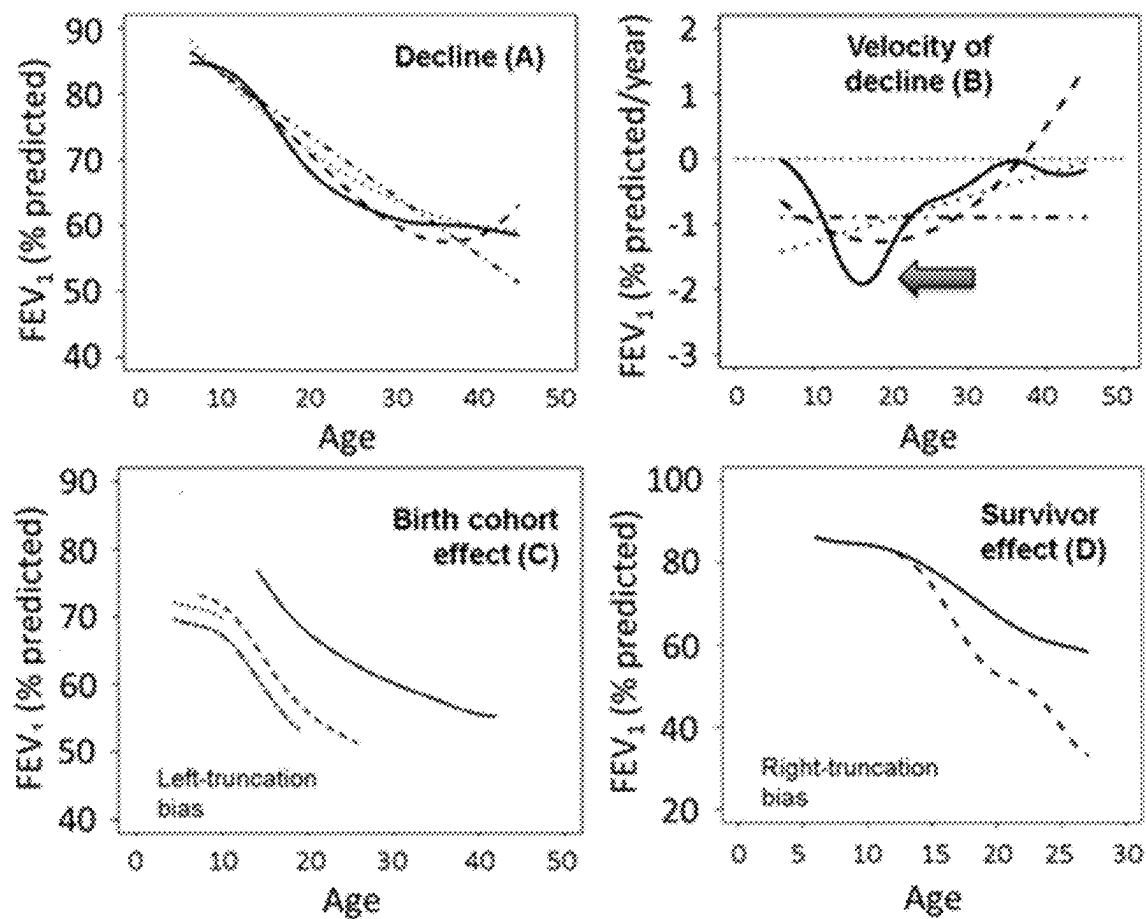
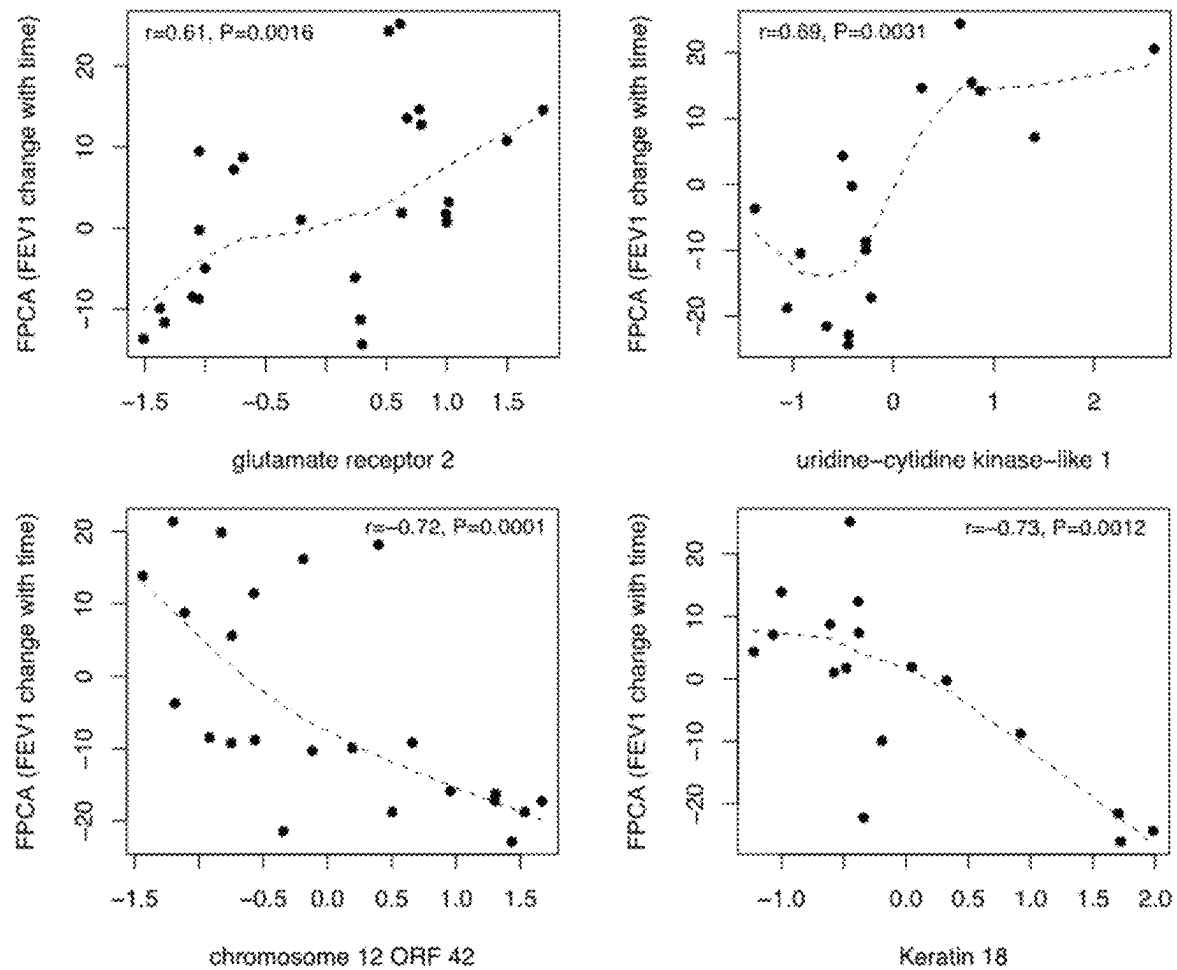


FIG. 4

**FIG. 5**

**FIG. 6**

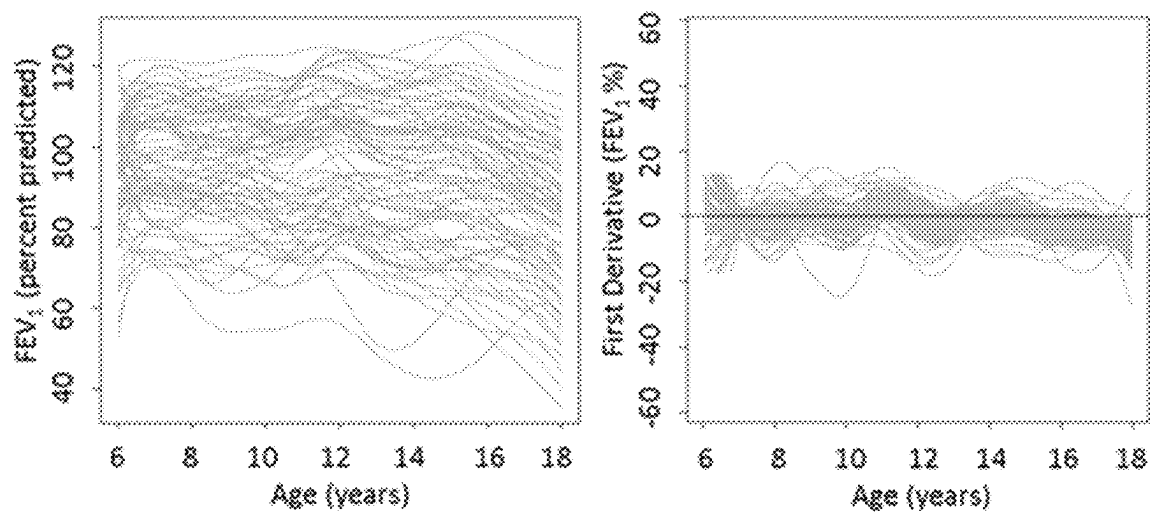


FIG. 7

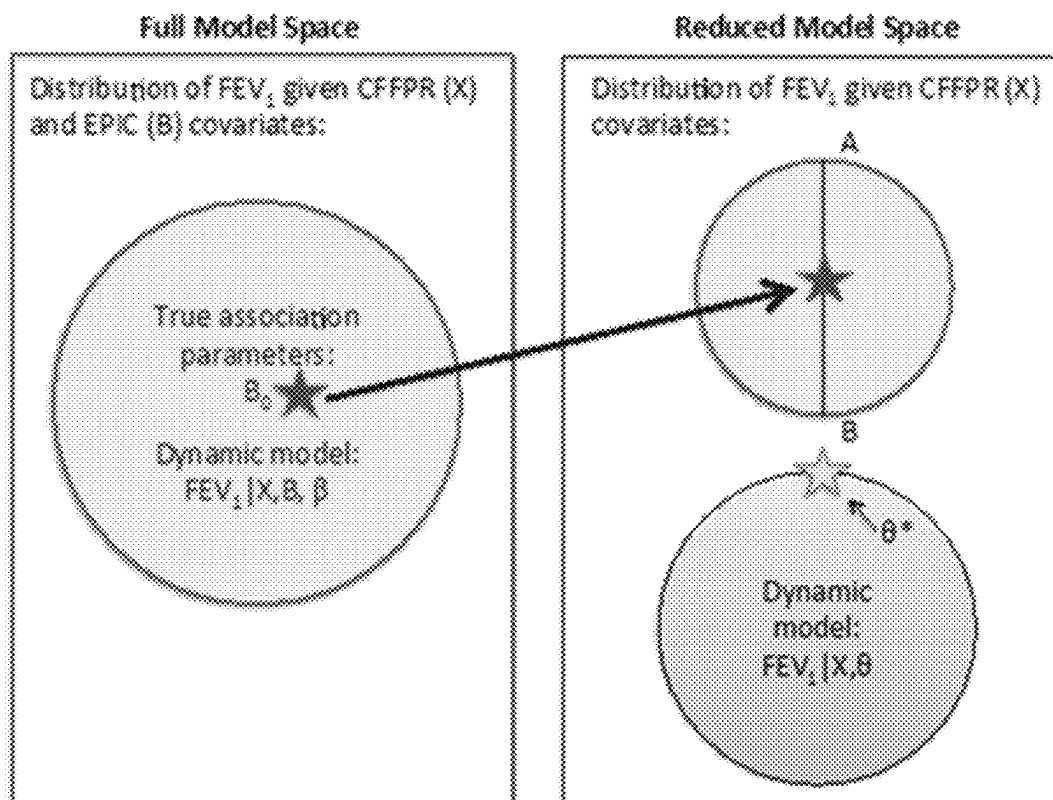
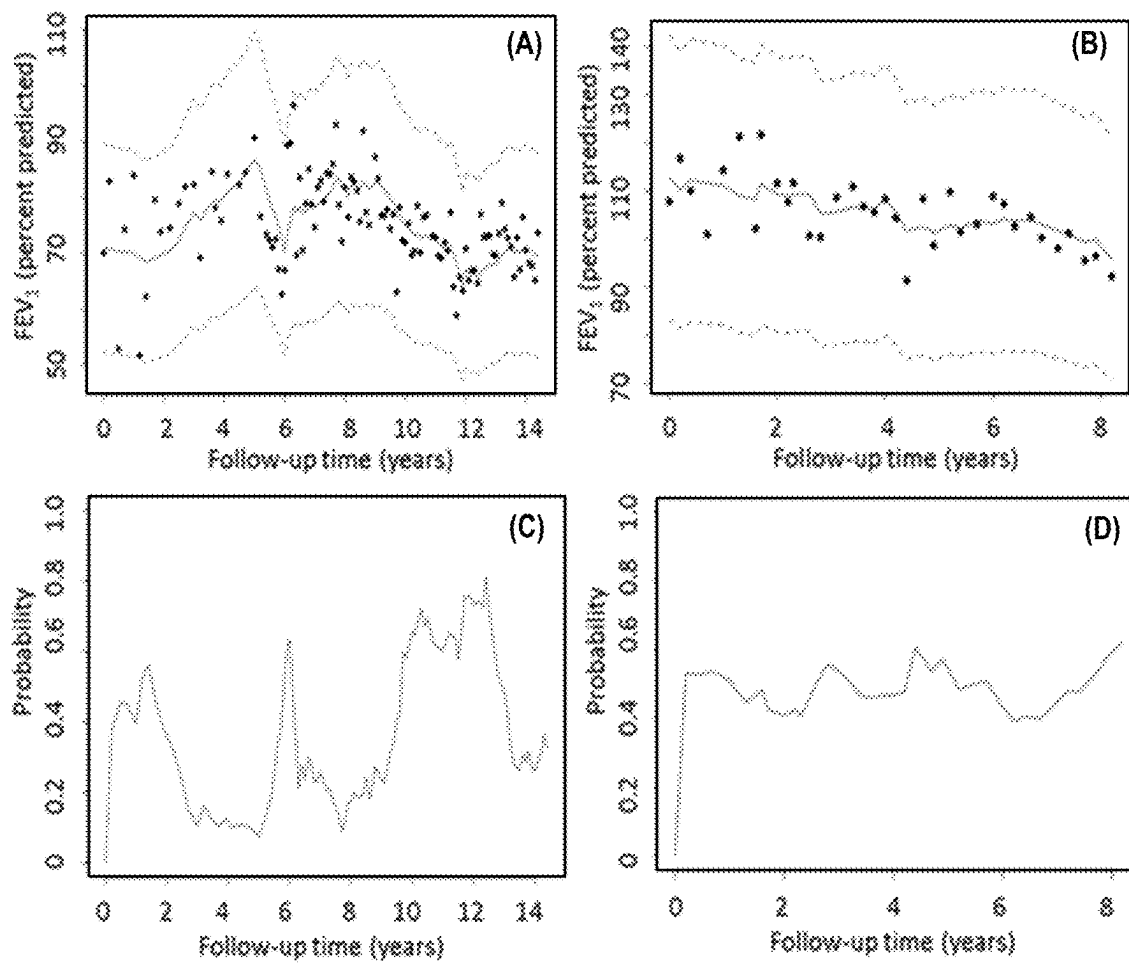


FIG. 8

**FIG. 9**

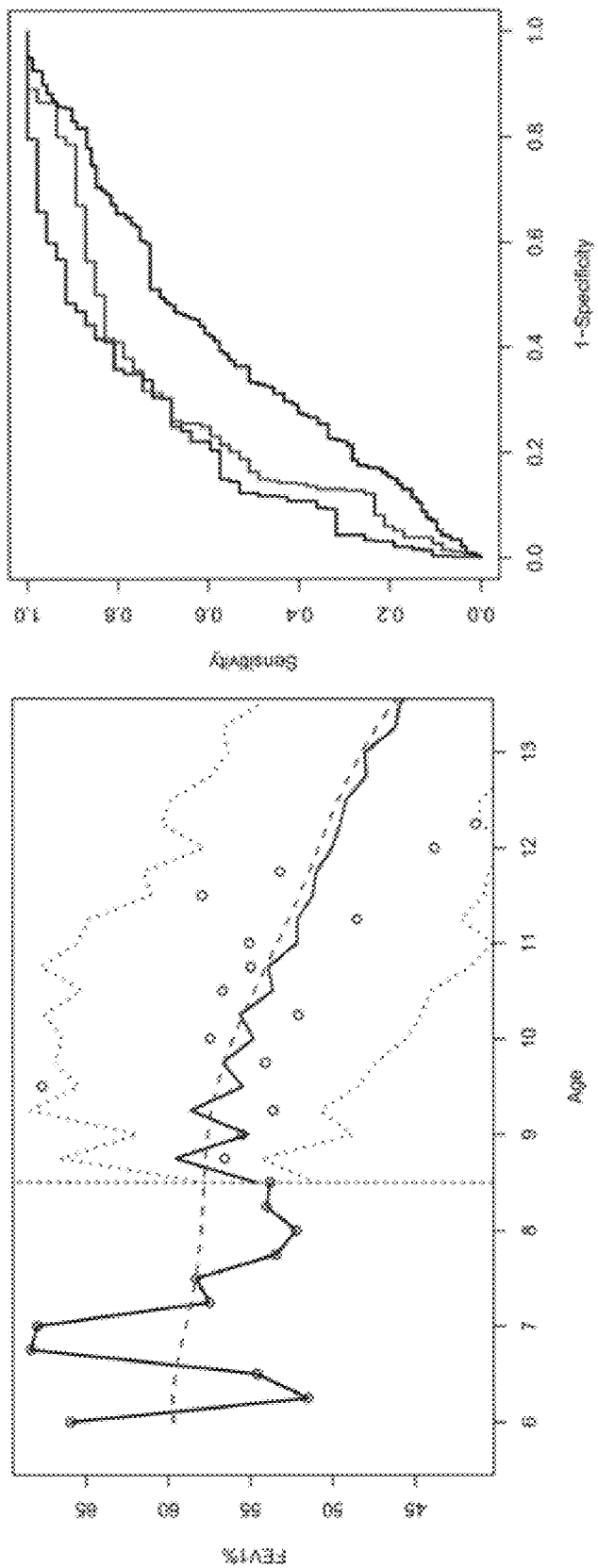


FIG. 10

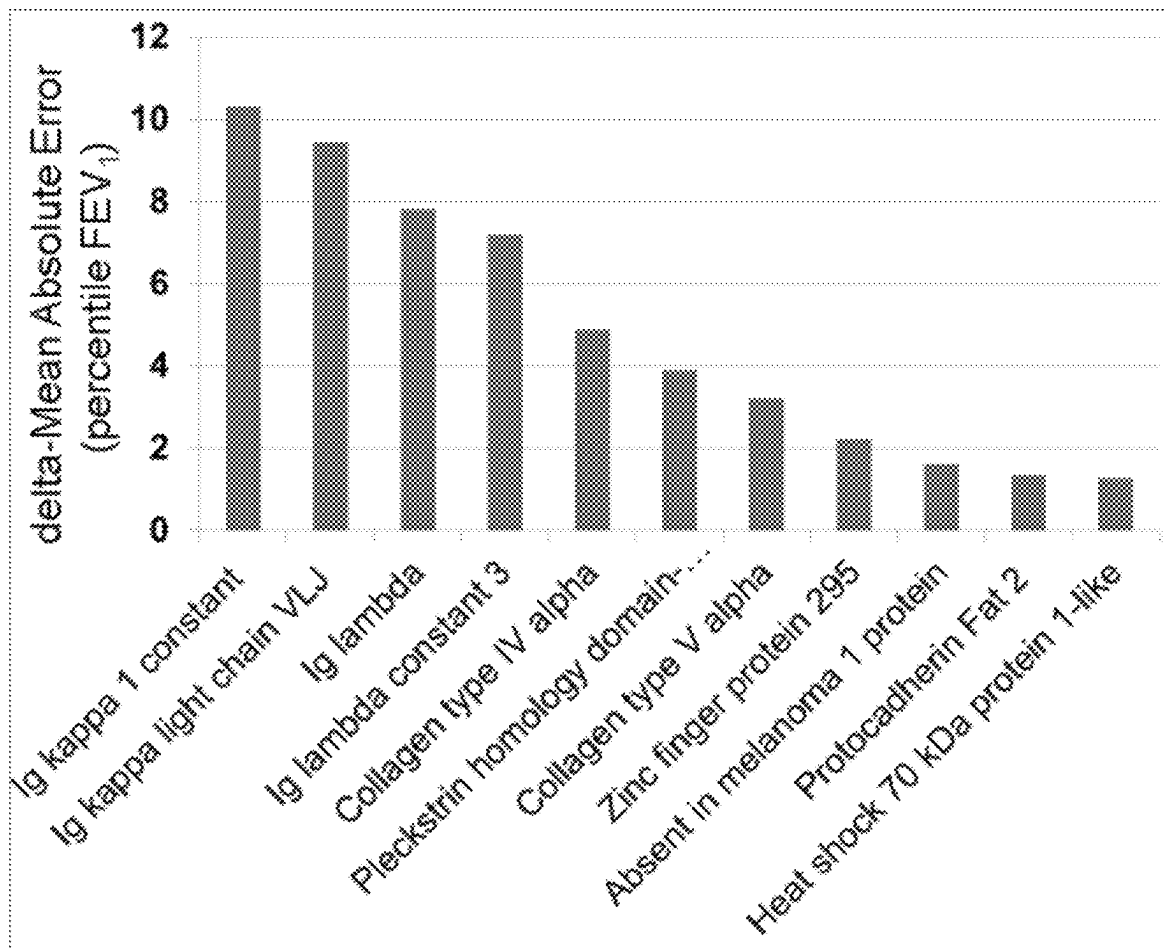


FIG. 11

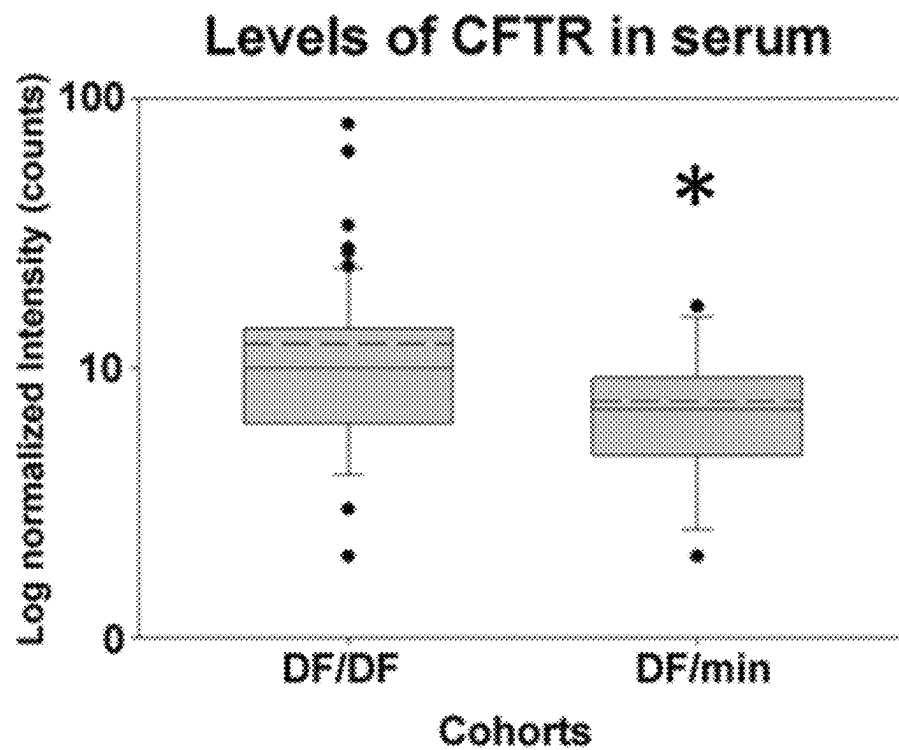


FIG. 12

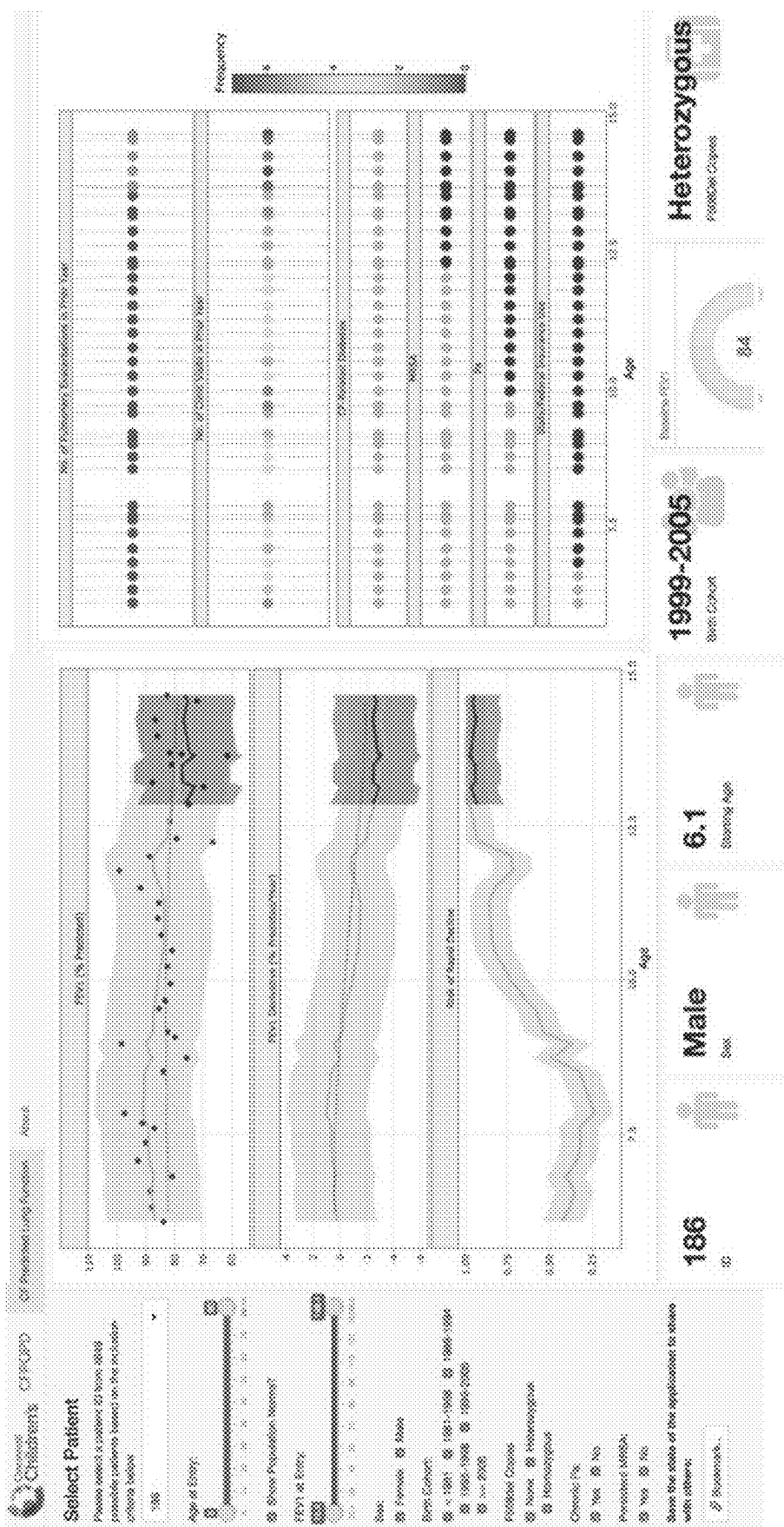


FIG. 13

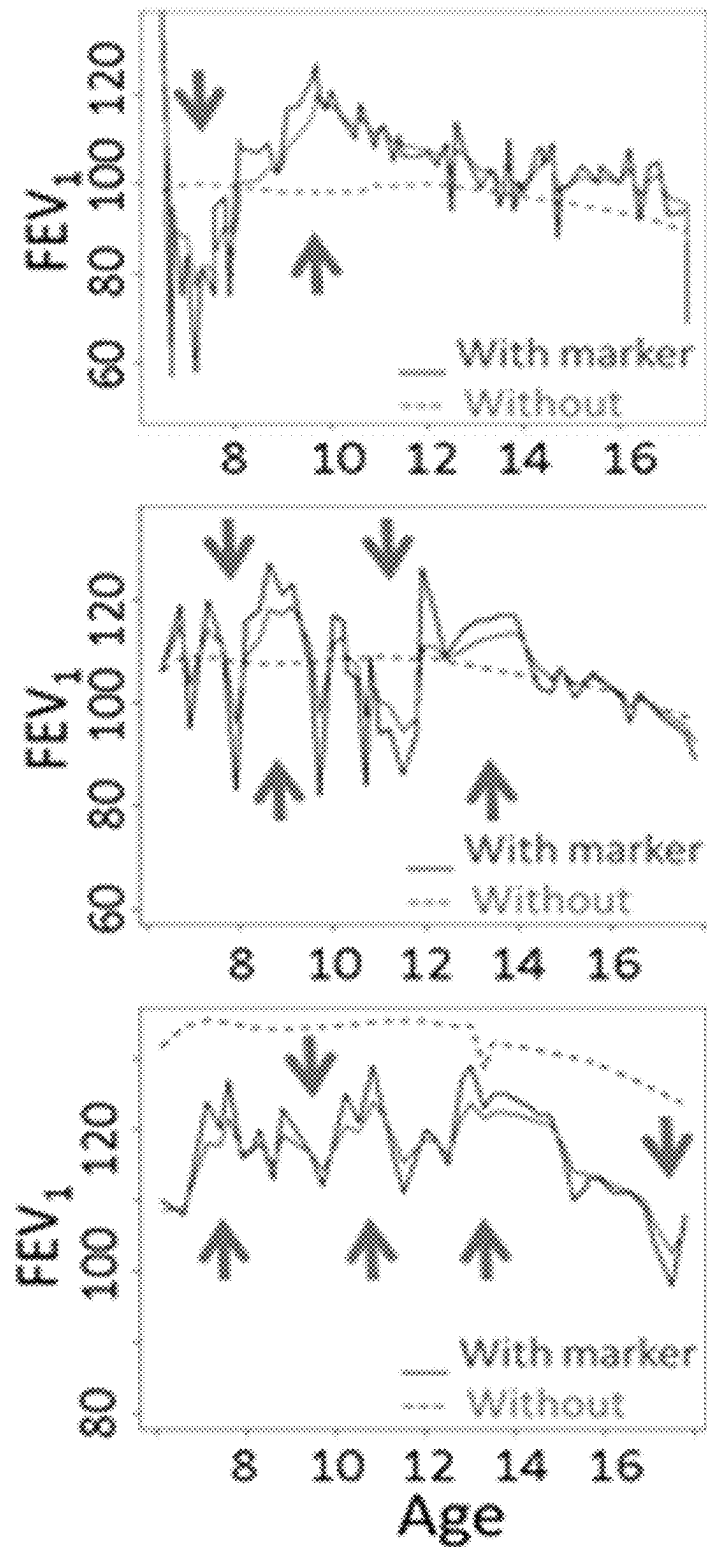
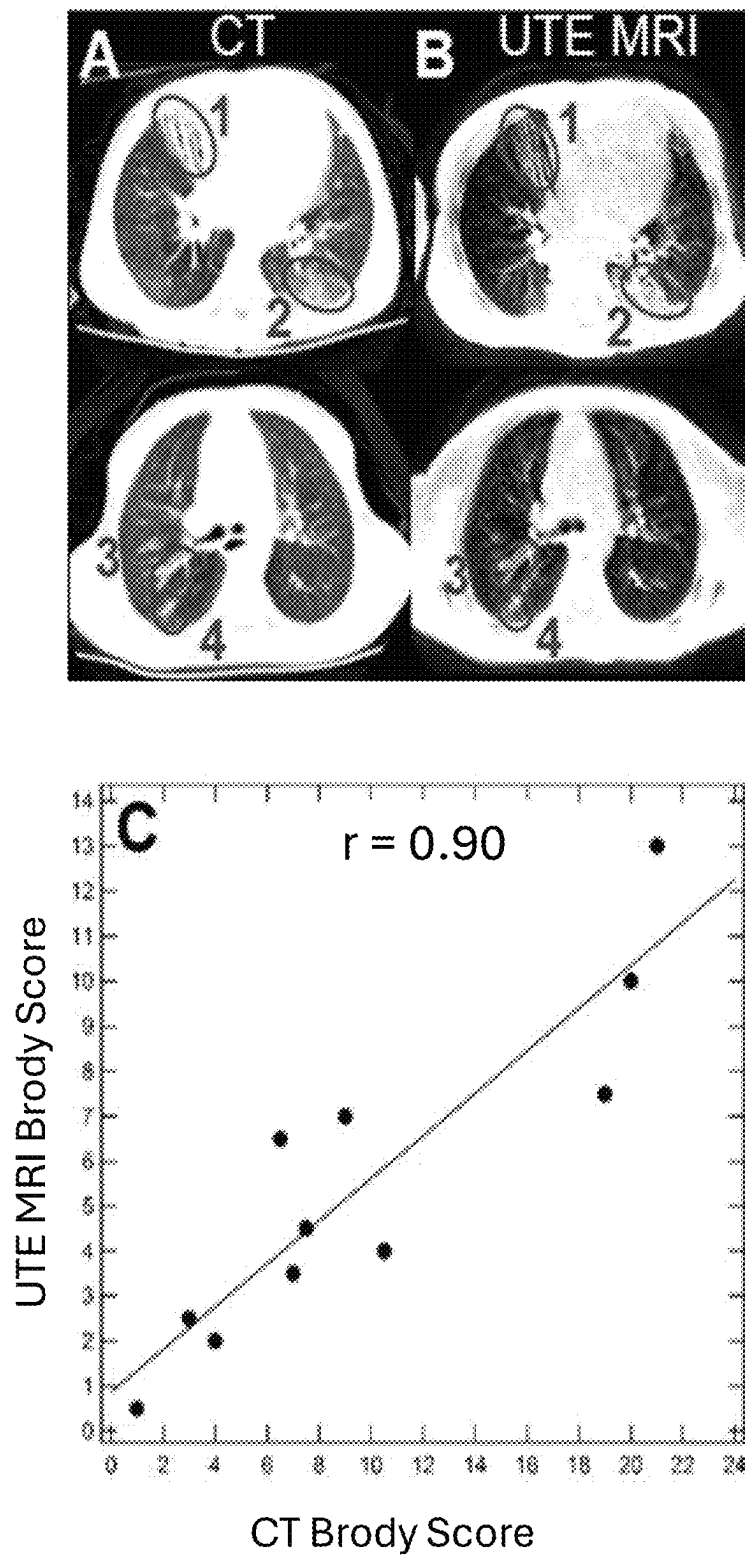


FIG. 14

**FIG. 15**

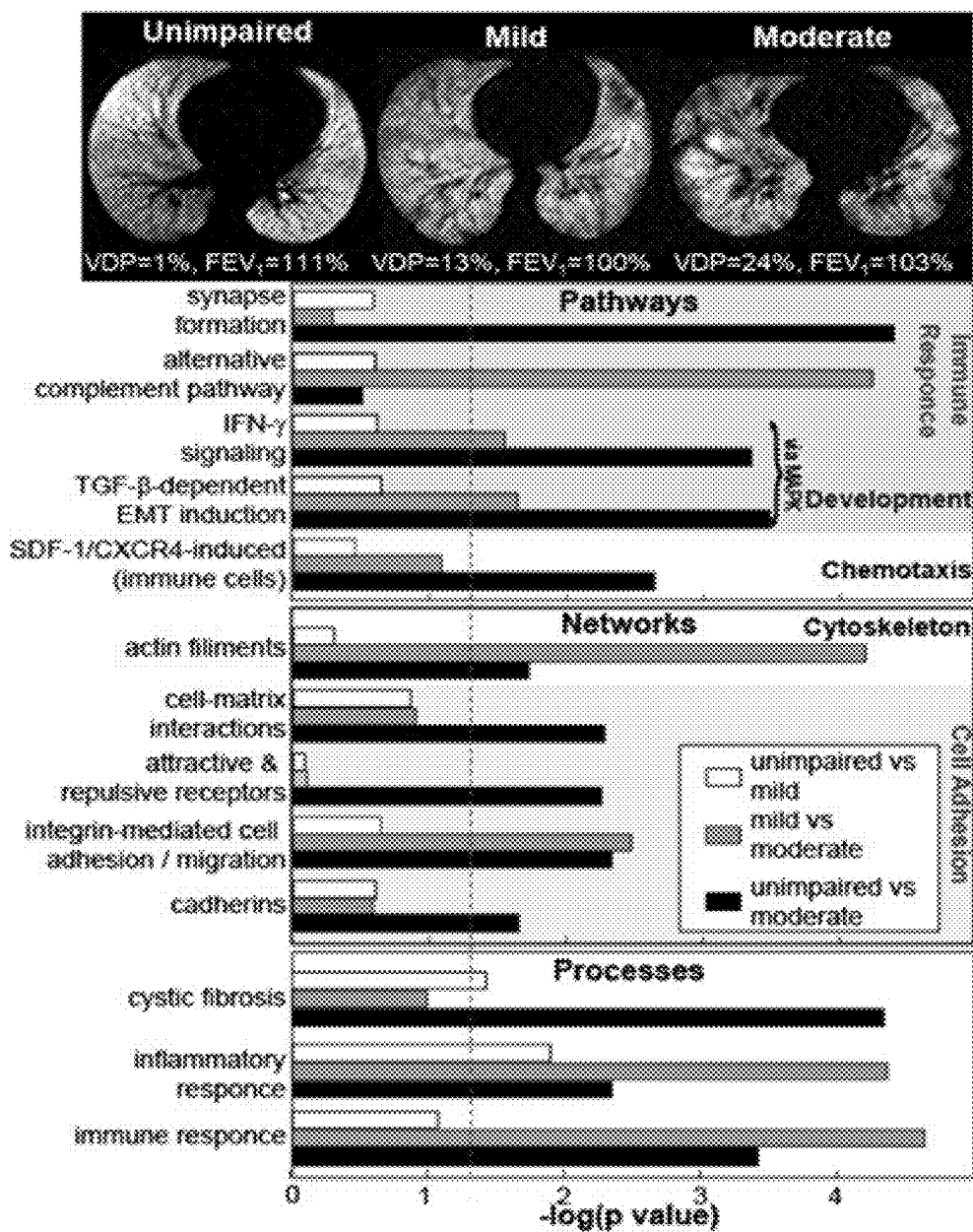


FIG. 16

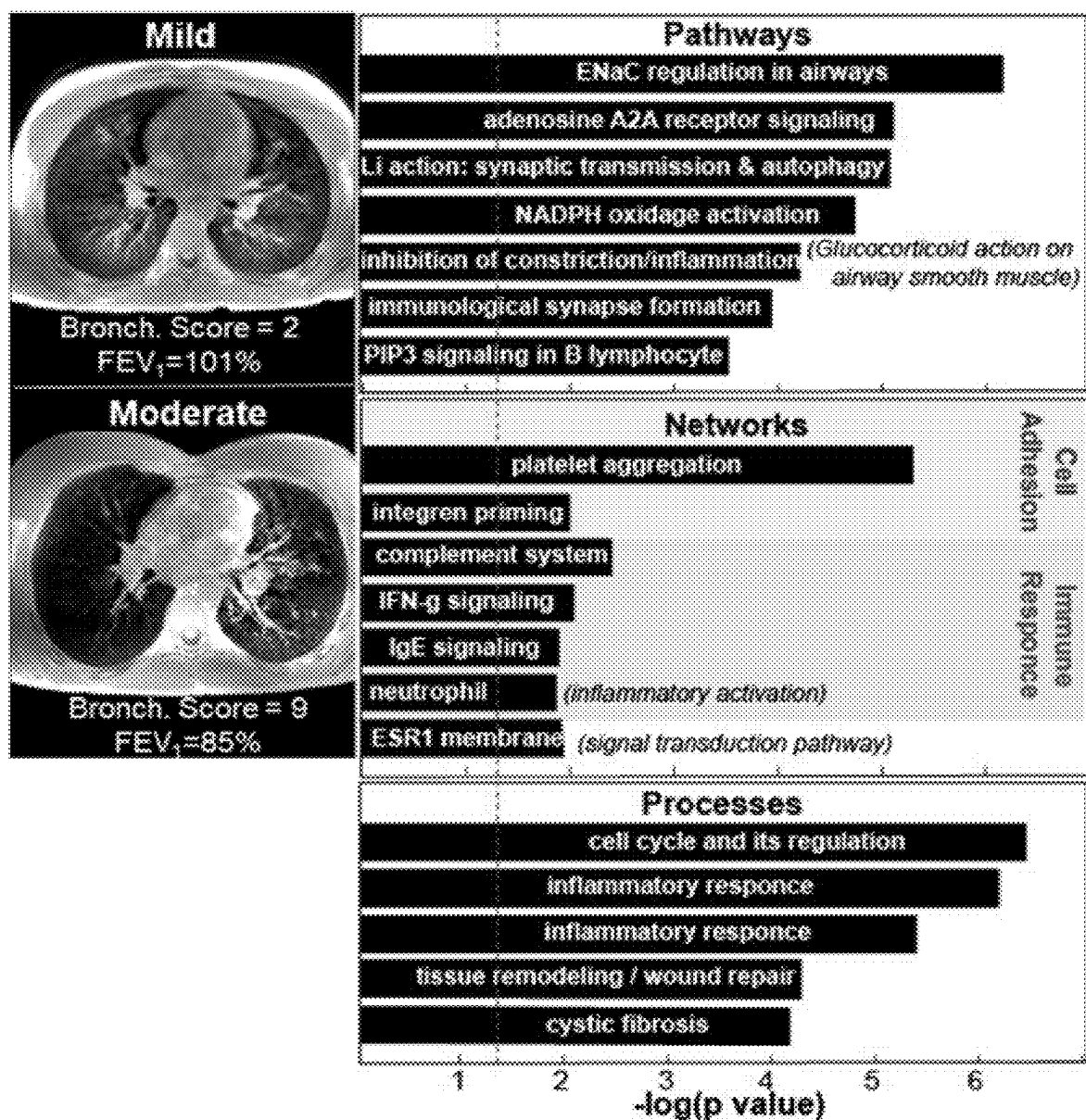


FIG. 17

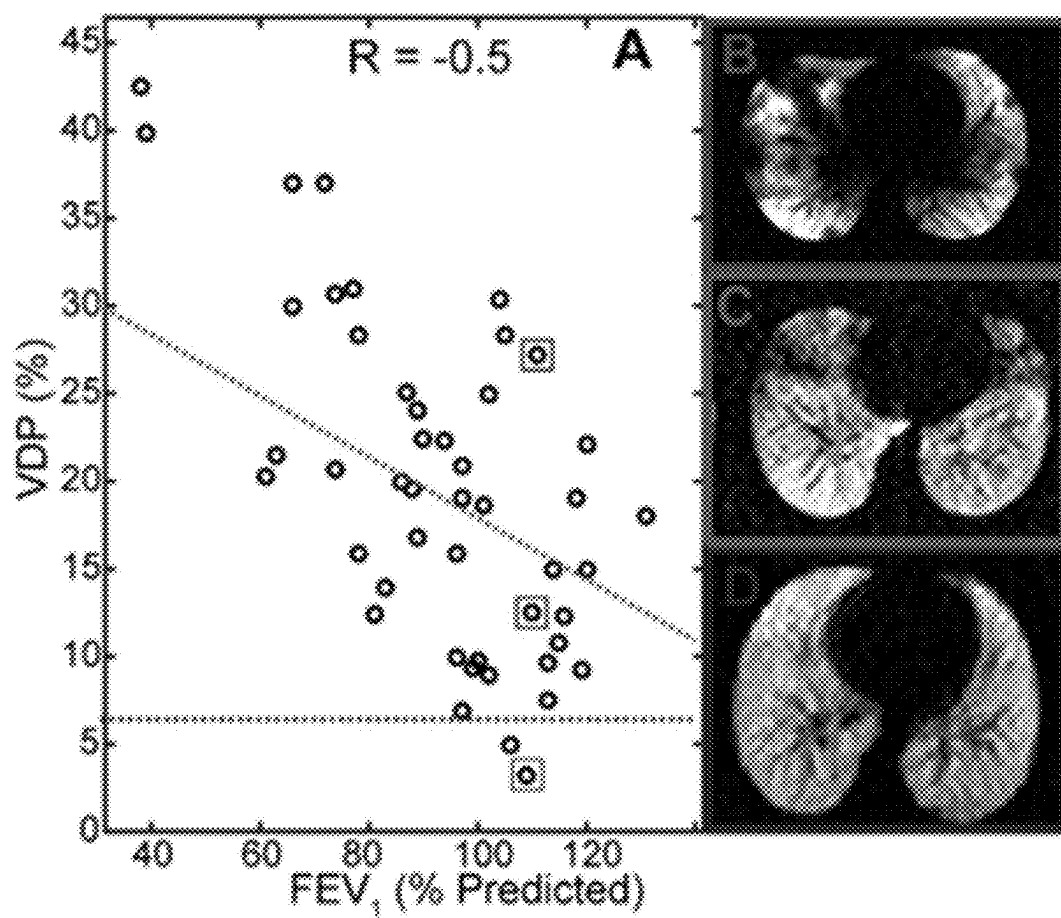


FIG. 18

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COMPOSITIONS AND METHODS FOR TREATMENT OF LUNG FUNCTION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. Ser. No. 16/922,119 filed Jul. 7, 2020, which claims priority to U.S. Ser. No. 15/927,575 filed Mar. 21, 2018, which claims priority to and benefit of U.S. Provisional Application No. 62/474,739, filed Mar. 22, 2017, the contents of each are incorporated in their entirety for all purposes.

GOVERNMENT SUPPORT CLAUSE

This invention was made with government support under HL142210, HL154105, HL116226 and HL125954 awarded by the National Institutes of Health. The government has certain rights in this invention.

BACKGROUND

Cystic fibrosis (CF) is an autosomal recessive disorder that affects approximately 30,000 individuals in the United States. The primary defect results from mutations of the cystic fibrosis transmembrane conductance regulator gene, which codes for the CFTR chloride channel. The protein is expressed predominantly on the apical surface of epithelial cells throughout the body (although low level expression has been detected in other tissues). Over 2,000 disease causing mutations have been identified in the CFTR gene, with the majority of patients (~90%) exhibiting at least one allele with the F508del mutation. Disease causing mutations fall into 5 classifications that result in abnormal CFTR protein that is either truncated, misprocessed/mislocalized, lacking channel gating function, or malformed due to improper gene splicing. With advances in new-born and other screenings, CF is usually diagnosed at birth. Although the determinants of disease are well characterized, forecasting disease progression has been extremely difficult and as of yet unsuccessful.

Care for CF patients has advanced rapidly over the past two decades, with an increase in patient longevity and quality of life that is unprecedented. The reasons for these improvements include a number of factors. First, the Cystic Fibrosis Foundation (CFF) has been tracking outcomes for nearly 50 years through their robust Patient Registry (CFF-PR), which includes patient data from nearly all CF patients in the US (individuals receiving care at accredited US CF centers), allowing assessment of outcomes and treatment responses. Next, there have been dramatic advances in new CF therapeutics (e.g. the development of recombinant human DNase, inhaled antibiotics including dissolved and dry powder tobramycin, aztreonam, hypertonic saline, low dose azithromycin to control inflammation, FDA approval of standardized pancreatic enzyme replacement, and most recently genotype-specific CFTR modulators such as KALYDECO® (4) and ORKAMBI® (5), and more recently TRIKAFTA® (6). In tandem with these new treatments, there has been a focus on the development of CF care guidelines and standardization of care across accredited CF care centers. This has helped to 'raise all boats' in the CF care community, accompanied by center-specific data to drive local quality improvement. Finally, understanding of disease severity predictors has advanced significantly, including the importance of weight in predicting pulmonary stability, the contribution of chronic *Pseudomonas* and

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MRSA infection to pulmonary decline and mortality, and the relationship between poorly controlled diabetes and disease progression. Indeed, these advancements have increased the median survival of CF patients to 41 years (CFF-PR—2014), and nearly 50% of CF patients alive today are adults. However, despite this impressive progress in CF care, lung function decline continues even in patients being treated with the best modulators of CFTR currently available. (See, e.g., Nataliya Volkova et al., Disease progression in patients with cystic fibrosis treated with ivacaftor: Data from national US and UK registries. JCF. DOI: 10.1016/j.jcf.2019.05.015.)

Accompanying the improvements in CF outcomes are a number of challenges that urgently require attention. There have been dramatic global improvements in the CF disease trajectory, but many patients have not fully benefited from the advancements described above. (FIG. 3.) Indeed, the average age of death of CF patients in a given year has remained remarkably static, with most patients dying of CF lung disease during their third decade of life (CFF-PR 2014 statistics). The burden of care also remains a significant challenge, as most adolescents and young adults need to spend approximately two hours daily dedicated to therapies that maintain health. Adherence to complex care regimens is often untenable, and this has led to the need to 'personalize' care such that patients commit to daily therapies that are most likely to benefit them individually. These commitments increase during periods of instability, and treatment of PE remains highly interruptive to daily care and negatively impacts quality of life. They also are sentinel markers of disease progression, as 25% of CF patients fail to recover lung functions following PEs. The benefits of care improvements have also challenged the capacity to monitor CF lung disease and CF manifestations in other organs. Assessing the relative benefit of new therapies in the context of relatively normal lung function (as measured by routine spirometry) is particularly challenging for CF providers. This requires the development of more sensitive tools to identify subjects most likely to benefit from various interventions and to monitor the impact of new therapies added to care plans. Finally, the conduct of clinical trials to advance CF outcomes and interventions can no longer rely on standard outcome measures such as forced expiratory volume in 1 sec (FEV₁), as excessively large and/or long clinical trials are needed to demonstrate improvements in crude measurements such as lung function. Thus, the CF field stands at a crossroads, where the benefits of the past limit the capacity to advance therapies and personalize care when relying upon standard measures of disease status. The disclosed methods, in certain aspect, may be used to address these gaps, seeking to produce vertical advancement in disease monitoring and prediction through the use of advanced biostatistical modeling of lung function, which may be further enhanced via coupling with novel molecular biomarkers and/or imaging. The disclosed methods may be used to identify those patients most likely to benefit from various interventions, and allow clinicians to monitor responses to precise and personal interventions.

The natural history of the disease is well studied; but disease progression is not well understood. Pulmonary decline typically begins in adolescence, but current measures tend to follow rather than predict outcomes. For example, if a marker predicted disease instability and erratic swings in lung function, established or novel interventions to prevent decline could be implemented. Several clinical measures track disease progression, including FEV₁, body mass index (BMI) and pulmonary exacerbations (PE). Pres-

ently, intervention is driven by lagging indications of lung function decline, which is far less beneficial than intervening in at risk subpopulations before decline is manifest. Therefore, methods that can be used to predict CF disease progression are highly desirable, as they would preemptively identify those at risk of future disease progression, allowing caregivers to tailor treatments and select intervention to prevent pulmonary decline. Personalizing therapy is a critical need in CF, as broad application of all available therapies leads to a high daily treatment burden and poor adherence. These measures are lagging indicators of disease progression that result from molecular changes directly or indirectly related to CFTR dysfunction. Furthermore, current monitoring of lung function data is inadequate, and fails to utilize novel biostatistical tools to identify patients at risk for future decline. The disclosed methods address one or more of aforementioned needs in the art.

BRIEF SUMMARY

Disclosed herein are methods for treating an individual at risk for non-linear lung function decline. The methods, in certain aspects, include the steps of a) determining one or more covariates associated with lung function in said individual, said covariate being selected from one or more of a clinical measure, a biomarker or an imaging marker; b) calculating a risk probability score based on said determining of one or more covariate, said risk probability score being used to characterize an individual as having no predicted lung impairment, mild predicted lung impairment, moderate predicted lung impairment, or severe predicted lung impairment; and c) treating said individual characterized as having mild predicted lung impairment, moderate predicted lung impairment, or severe predicted lung impairment with one or more of increased frequency of disease monitoring, increased frequency of infection monitoring, an anti-inflammatory therapy, or combinations thereof.

BRIEF DESCRIPTION OF THE DRAWINGS

Those of skill in the art will understand that the drawings, described below, are for illustrative purposes only. The drawings are not intended to limit the scope of the present teachings in any way.

FIG. 1. Lung function in a typical CF patient. Lung function is separated into stages, with a stable period (light grey), an erratic period with large swings in FEV₁ (dark grey), and a decline stage (black).

FIG. 2. Disease progression in CF. In depth serum proteomic analysis can capture molecular changes in serum that give rise to downstream organ pathology.

FIG. 3. Lung function decline with age. Comparison from data compiled from 1990-2014. Despite improvement in lung function at early age, the rate of decline is unaffected by modern therapies. Source: CF Foundation Patient Registry data.

FIG. 4. Box plots of CF disease severity serum biomarkers. Example of differential expression identified serum biomarkers of lung disease severity. Forty-four candidates whose quantities differ by 0.35-5.4-fold between mild and severe disease have been identified.

FIG. 5. Functional Data (FD) analysis model is superior to presently available models of lung function decline. CFFPR analyses with point estimates from FD analysis (solid line), cubic (dashed line), quadratic (dotted line) and conventional linear (dot-dash line) mixed models of decline (A) and rate of decline (B); stratified by birth cohorts born before

1981 (solid line), 1981-1988 (dashed line), 1989-1994 (dot-dash line), and after 1994 (dotted line) (C); stratified by survival (solid line) and death (dashed line) for patients less than 19 years of age (D). The FD model in (A) reflected the dynamic status of lung function during young adolescence/early adulthood; traditional models found no changes and even indicated gains in lung function (e.g. cubic). The arrow at the "dip" in (B) shows 1) Patients attained most rapid decline at median age (IQR) 16.3 (13.5, 21.0) years; 2) Degree of maximal FEV₁₀% loss was variable (mean: 1.98% pred/yr, 95% CI: 1.86, 2.10). Longitudinal FEV₁₀% measures shared correlation for up to 9 years, highlighting the potential of short-term clinical interventions to impact long-term lung function. Further subgroup analysis revealed that FEV₁₀% curves vary according to survivorship and birth cohort (C-D), highlighting left- and right-truncation biases, respectively.

FIG. 6. Biomarker correlation with functional principle component (FPCA) analysis of lung function. Examples shown for 4 of 18 biomarkers discovered in preliminary cross-sectional proteomic analyses that significantly correlate with FPCA analysis of FEV₁. The first principle component score (FPC1, y-axis); the thick dashed black line is the fit to the data using a scatterplot smoother; positive association (r) indicates that higher values of these markers may lead to worsening FEV₁ trajectory, while negative associates (−r) indicates that higher values of these markers correspond to improvements in the FEV₁ trajectory. Although these studies were conducted in discovery mode to capture the maximal number of biomarkers, data on individual markers may be improved under targeted MS and ELISA analyses.

FIG. 7. On the left, smoothed FEV₁ observed over age (in years) for the EPIC cohort. On the right, the corresponding rate of change in FEV₁ (expressed as % predicted/year) over age.

FIG. 8. Schematic of biomarker integrated dynamic modelling of FEV₁. On the left, the full model space for all CF patients and possible covariates; the large circle is the space for the full dynamic prediction model, and the star denotes the true associates between each CFFPR/EPIC covariate and rapid (non-linear) decline. On the bottom right, the bottom circle is the space for the external data model that is fit to the CFFPR; the bottom star marks the parameters Θ^* that would minimize the mathematical distance between the full model with all covariates and the model with only the CFFPR covariates. On the upper right, the circle shows a star being mapped to the same space as the external model, in order to perform model calibration and obtain more efficient, unbiased estimates.

FIG. 9. Dynamic prediction modeling. Dynamic predictions for female (left column) and male (right column) CF patients. In the left column, the female patient had data on 111 encounters in the CFFPR; her age at entry (follow-up time: 0 years on the axis) was 6.1 years. Her decline in FEV₁ was highly variable over time (A) with probability of rapid decline on the y-axis in (B); periods of increased risk of rapid (non-linear) decline shown at 16-18 years of age (follow-up time: 10-12 years on the x-axis). By contrast, the male patient (right column) had steadier decline and less availability (C). He had data on 35 encounters in the CFFPR; his age at entry was 14.5 years. He had decreased risk of rapid (non-linear) decline (D), compared to the female patient. Rapid decline was defined as a rate of change in longitudinal FEV₁ that fell below 1.5% predicted/year. Threshold was determined as previously reported (53).

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FIG. 10. FEV₁ and clinical event forecasting performance. Left panel: the FEV₁ forecasting performance for a representative subject from the CFF-PR. Observed FEV₁ data (black dots) is fitted with the model (black line) over age; the model was used to predict his subsequent FEV₁ data shown after the vertical bar in the plot. To the right of the vertical bar, the solid line and dashed lines are the predicted FEV₁ and corresponding 95% confidence bands. The dotted line is the smooth, population-level trajectory for FEV₁. Right Panel: the ROC curves for the model fit to PE data. The upper curve is the proposed model, which has the highest AUC (0.74), followed by a model with no subject-level effect and logistic regression (0.68 and 0.61, respectively).

FIG. 11. Marker improvement of FEV₁ prediction. Protein markers in serum improve the ability to predict future lung function decline, reducing the error in prediction of absolute FEV₁ by up to greater than 10 percentile points.

FIG. 12. Quantitation of CFTR by tandem mass spectrometry. Fractionated serum is analyzed for levels of CFTR in homozygote F507del patients versus those with 1 F508del allele with the other allele producing minimal protein. Solid line is median. Dashed line is average. DF/DF average $\pm$ SEM=12.34 $\pm$ 1.44 vs DF/min average $\pm$ SEV=7.5 $\pm$ 1.09. * connotes significant difference, p=0.0092.

FIG. 13. GUI Prototype. The GUI (graphical user interface) was composed using the R statistical programming language, specifically using the flexdashboard, shiny, and plotly packages. See R Core Team (2017). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL <https://www.R-project.org/>; Barbara Borges and J J Allaire (2017). flexdashboard: R Markdown Format for Flexible Dashboards. R package version 0.5.1. <https://CRAN.R-project.org/package=flexdashboard>; Winston Chang, Joe Cheng, J J Allaire, Yihui Xie and Jonathan McPherson (2017). shiny: Web Application Framework for R. R package version 1.0.5. <https://CRAN.R-project.org/package=shiny>; Carson Sievert, Chris Parmer, Toby Hocking, Scott Chamberlain, Karthik Ram, Marianne Corvellec and Pedro Despouy (2017). plotly: Create Interactive Web Graphics via 'plotly.js'. R package version 4.7.1. <https://CRAN.R-project.org/package=plotly>.

FIG. 14. Lung function prediction with biomarkers. FEV₁ trajectory is predicted for three CF patients with proteomic biomarker data from 87 subjects. Clinical data is shown in (black), while the algorithm's prediction is shown without (dashed) or with (solid) or 3 of the top 20 markers (coded to maintain blind during ongoing studies to develop the algorithm for clinical translation). Arrows demonstrate increased accuracy for predicting lung function, especially for upward or downward deflections from the smoothed trajectory. Adding markers to the model significantly improved personalized prediction of lung function trajectories, demonstrating compelling evidence that the disclosed approach can translate into a superior tool for CF.

FIG. 15. CF pathologies (rd circles) on CT (A) and ultra-short echo-time (UTE) magnetic resonance imaging (MRI) (B) in 1-yr old patients: (1) bronchiectasis, (2) ground glass opacity, (3) wall thickening, and (4) mucus plugging. (C) "Brody" scoring for both modalities.

FIG. 16. Ventilation vs. proteomics. Top) CF patients with FEV₁>100% and no, mild, and moderate ventilation impairment via ¹²⁹Xe MRI. Bottom) Proteomic differences in CF

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patient plasma collected on the day of MRI. Cohort Comparisons are indicted in the legend (center). Vertical line: p=0.05.

FIG. 17. Structural UTE MRI vs. proteomics. Left) CF patients with mild (top) and moderate (bottom) remodeling via UTE. Right) Proteomic differences between patients with mild and moderate remodeling. Vertical line: p=0.05.

FIG. 18. ¹²⁹Xe MRI of lung dysfunction. A) ¹²⁹Xe VDP vs FEV₁ in CF patients. Mean VPD of controls is shown (blue line). While FEV₁ and VDP correlate (red line, p<0.001), scatter is seen in CF patients with normal FEV₁. B-D) Ventilation images of CF patients with FEV₁>100%. These patients display different regional ventilation, ranging from normal (D, VDP<4%) to impaired (B, VDP=28%), indicating the higher sensitivity of ¹²⁹Xe MRI.

DETAILED DESCRIPTION

Definitions

Unless otherwise noted, terms are to be understood according to conventional usage by those of ordinary skill in the relevant art. In case of conflict, the present document, including definitions, will control. Preferred methods and materials are described below, although methods and materials similar or equivalent to those described herein can be used in practice or testing of the present invention. All publications, patent applications, patents and other references mentioned herein are incorporated by reference in their entirety. The materials, methods, and examples disclosed herein are illustrative only and not intended to be limiting.

As used herein and in the appended claims, the singular forms "a," "and," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a method" includes a plurality of such methods and reference to "a dose" includes reference to one or more doses and equivalents thereof known to those skilled in the art, and so forth.

The term "about" or "approximately" means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, "about" can mean within 1 or more than 1 standard deviation, per the practice in the art. Alternatively, "about" can mean a range of up to 20%, or up to 10%, or up to 5%, or up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 5-fold, and more preferably within 2-fold, of a value. Where particular values are described in the application and claims, unless otherwise stated the term "about" meaning within an acceptable error range for the particular value should be assumed.

As used herein, the term "effective amount" means the amount of one or more active components that is sufficient to show a desired effect. This includes both therapeutic and prophylactic effects. When applied to an individual active ingredient, administered alone, the term refers to that ingredient alone. When applied to a combination, the term refers to combined amounts of the active ingredients that result in the therapeutic effect, whether administered in combination, serially or simultaneously.

The terms "individual," "host," "subject," and "patient" are used interchangeably to refer to an animal that is the object of treatment, observation and/or experiment. Generally, the term refers to a human patient, but the methods and

compositions may be equally applicable to non-human subjects such as other mammals. In some embodiments, the terms refer to humans. In further embodiments, the terms may refer to children.

The standard of care for patients with cystic fibrosis (CF) has advanced rapidly but CF patients still suffer routinely from pulmonary exacerbations (PE) and these are under reported. Current treatment methods are reactive to drops in lung function. A prognostic test that may be used to predict lung function decline in CF patients based on the combination of clinical data with molecular and imaging markers may be used to proactively treat CF patients. Variables such as historic lung function measures, biomarkers in blood, and regional lung structural and functional information obtained by imaging may be combined into a functional data algorithm that may be used to forecast the projection of lung function trajectory for a time period, for example, about 6 to about 12 months. Spirometric measures of lung function, namely the forced expiratory volume in one second (FEV₁), remain the most commonly used clinical measure of disease progression. It may be noted that FEV₁ percentile (standardized to CF) is not equivalent to FEV₁ percent predicted (standardized to general population). As used herein, reference to FEV₁ refer to FEV₁ percentile or FEV₁ percent predicted.

However, these measures provide only whole-lung metrics of lung function, making them insensitive to early disease. As a result of this inherent insensitivity and improved clinical care, the majority of pediatric CF patients now have normal FEV₁, making it increasingly difficult to detect early lung disease, when intervention would be most beneficial. As clinicians move toward maintaining higher lung function for longer periods of time, it will become increasingly important to develop more sensitive measures of CF lung disease and/or more specific biomarkers. Further, combining these non-invasive testing modalities to predict exacerbations and lung function decline may be advantageous. The limitations of FEV₁ and the pressing need for higher sensitivity for forecasting lung disease progression are relevant to a number of disorders, including COPD, severe asthma, and non-CF bronchiectasis. Here, methods employing an algorithm to predict structural remodeling in, and function of, the CF lung are disclosed. Such methods

may be used with or without the use of one or more of the disclosed biomarkers, or sets of biomarkers disclosed herein.

In one aspect, Applicant has developed a dynamic prediction model to produce a novel diagnostic algorithm that identifies individuals at risk of lung function decline. In another aspect, Applicant has developed dynamic prediction modeling to identify CF patients who develop rapid pulmonary decline during adolescence. Disclosed herein are methods and assays which may be used in combination with an algorithm for prediction of CF lung disease progression that is superior to current practice. In one aspect, the disclosed methods may employ the use of novel biomarkers of disease severity that may enhance the ability of FD analysis to predict FEV₁ decline.

Three important measures are used in monitoring CF disease progression and response to therapy; FEV₁, PE frequency, and BMI. A rapid deterioration of lung function persists in CF patients, especially during adolescence. (FIG. 3). Although FEV₁, PE, and BMI are trailing indicators of disease progression, they are phenotypic culminations of molecular changes that persist and continue to influence the course of disease. In addition to the deleterious effects of CF disease progression (such as mucus plugging, bacterial infection, and inflammation), these clinical measures reflect a molecular basis of disease that may be detectable in biological specimens such as blood. Serum and/or plasma markers may be used to predict disease progression and may allow better estimation of future disease trajectory, which would inform therapy and facilitate development of patient-specific treatment regimens. Applicant has identified systemic changes reflected in the blood that culminate in measurable pulmonary decline as children enter adolescence, including serum and/or plasma markers that may be used to better estimate future disease trajectory, which may then be used to inform therapy and facilitate development of patient-specific treatment regimens.

In one aspect, described herein are novel methods for the prediction of lung function decline and treatment of individuals having or predicted to have lung function decline, such as individuals diagnosed with cystic fibrosis (CF).

In one aspect, a method of predicting lung function, more particularly, the likelihood of a decline in lung function, in an individual is disclosed. In certain aspects, the methods may include detecting expression of a one or more biomarkers selected from Table 1.

TABLE 1

Protein Isoform Name and Gene Name (Abbreviation)	
Protein isoform name	Gene name
Immunoglobulin alpha-1 heavy chain constant region	IGHA1
kappa 1 immunoglobulin constant	IGKC
Immunoglobulin kappa light chain VLJ region	IGK
immunoglobulin lambda	IGL
Immunoglobulin lambda constant 2	IGLC2
Immunoglobulin lambda constant 3	IGLC3
Tight junction protein 3	TJP3
Alpha-1-acid glycoprotein 2 precursor	ORM2
Signal-induced proliferation-associated 1-like protein 3	SIPA1L3
Plakophilin 4	PKP4
Inter-alpha (globulin) inhibitor H2	ITIH2
Absent in melanoma 1 protein	AIM1
Actin filament-associated protein 1 isoform X2	AFAP1
Serine/threonine-protein phosphatase PP1-beta catalytic subunit	PPP1CB
Retinol-binding protein 4	RBP4
Fermitin family homolog 1	FERMT1
Actin, cytoplasmic 2	ACTG1
Transthyretin	TTR
Melanoma inhibitory activity protein 3 isoform X2	MIA3
Pleckstrin homology domain-containing family G member 1 isoform X3	PLEKHG1

TABLE 1-continued

Protein Isoform Name and Gene Name (Abbreviation)	
Protein isoform name	Gene name
Pleckstrin homology domain-containing family A member 5 isoforms X1-X9	PLEKHA5
Pleckstrin	PLEK
Zinc finger protein 295	Zbtb21
Protocadherin Fat 2	FAT2
Cadherin-related family member 2 isoform X1	CDHR2
Voltage gated calcium channel alpha 1F subunit	CACNA1F
Actin filament associated protein	AFAP
Heat shock 70 kDa protein 1-like	HSPA1L
EF-hand calcium binding protein 2	EFCBP2
Polyamine modulated factor-1 (PMF1) protein	PMF1
Keratin 16	KRT16A
Keratin 18	KRT18
DNA replication ATP-dependent helicase/nuclease DNA2 isoform X4	DNA2
Non-lens beta gamma-crystallin like protein	CRYBG1
Bromodomain and WD repeat-containing protein 1 isoform X1	BRWD1
Collagen type V alpha 3 chain	COL5A3
Collagen type IV alpha	COL4A
Dynein heavy chain 12, axonemal isoform X5	DNAH12
Serine/arginine repetitive matrix protein 1	SRRM1
Ras-related protein Rap-1A	RAP1A
AT-hook-containing transcription factor isoform 1	AKNA1
Coiled-coil domain-containing protein 18 isoforms X1-X7	CCDC18
Coiled-coil domain-containing protein 180	CCDC180
Patatin-like phospholipase domain-containing protein 2 isoform X1	PNPLA2

In one aspect, predicting lung function in an individual diagnosed with cystic fibrosis is disclosed. In certain

aspects, the methods may include detecting expression of a one or more biomarkers selected from Table 2.

TABLE 2

Panel of protein biomarker covariates for algorithm that improve algorithm performance in FEV₁ predictions over a period from 3 weeks to 4 months after biomarker collection.
Protein name (gene name)

1. Alstrom syndrome protein 1 isoform 1
2. aminopeptidase O isoform X9 (C9orf3)
3. ankyrin-3 isoform 3
4. ARAP2 protein (ARAP2)
5. Bifunctional glutamate/proline—tRNA ligase (EPRS)
6. C10ORF6
7. CECR2 protein (CECR2)
8. cingulin-like 1 (CGNL1)
9. coiled-coil domain-containing protein 93 isoform X3 (CCDC93)
10. complement component C2 (C2)
11. complement component C6 isoform X7 (C6)
12. complement factor H (CFH)
13. connector enhancer of kinase suppressor of ras 2 isoform X2 (CNKSR2)
14. CTP:phosphocholine cytidyltransferase
15. DNA annealing helicase and endonuclease ZRANB3 (ZRANB3)
16. EPRSN1
17. fibrinogen alpha chain preproprotein, isoform alpha (FGA)
18. filamin 2 (FLN2)
19. Fras1-related extracellular matrix protein 2 (FREM2)
20. FYVE, RhoGEF and PH domain-containing protein 5 isoform X2 (FGD5)
21. gelsolin isoform X3 (GSN)
22. glutamate receptor-interacting protein 1 isoform X7 (GRIP1)
23. hCG1656772
24. histidine-rich glycoprotein isoform X1 (HRG)
25. interferon regulatory factor-2 binding protein 2B
26. junction-mediating and -regulatory protein isoform X2 (JMY)
27. KIAA0328 protein KIAA0328
28. la-related protein 1 isoform X5 LARP1
29. MAX gene-associated protein isoform X11 (MGA)
30. paraoxonase-3
31. Platelet glycoprotein Ib alpha chain (GP1BA)
32. PMFBP1 protein (PMFBP1)
33. PRO2841
34. protein arginine N-methyltransferase 3 isoform X1 (PRMT3)
35. Protein SCAF11 (SCAF11)
36. ral GTPase-activating protein subunit alpha-1 isoform X12 (RALGAPA1)
37. Receptor-type tyrosine-protein phosphatase C (PTPRC)

TABLE 2-continued

Protein name (gene name)
38. RGS3 isoform C2PA-RGS3 (RGS3)
39. Rho-GTPase activating protein 10 (ARHGAP10)
40. RNA-binding motif protein, Y chromosome, family 1 member B isoform X1 (RBM1B)
41. protein-methionine sulfoxide oxidase MICAL3 isoform X10 (MICAL3)
42. Protein sidekick-2 (SDK2)
43. saccin isoform X3 (SACS)
44. SAPS domain family, member 2 (SAPS2)
45. serum arylalkylphosphatase precursor
46. spermatogenesis-associated protein 31C1 isoform X1 (SPATA31C1)
47. synaptotagmin-like protein 2 isoform X10 (SYTL2)
48. translation initiation factor (IF2)
49. triadin isoform X17 (TRDN)
50. uridine-cytidine kinase-like 1 isoform X7 (UCKL1)
51. utrophin isoform X4 (UTRN)
52. WD repeat-containing protein 64 isoform X2 (WDR64)
53. zinc finger and AT hook domain containing (ZFAT)

In one aspect, predicting lung function in an individual diagnosed with cystic fibrosis is disclosed. In certain

aspects, the methods may include detecting expression of one or more biomarkers selected from Table 3.

TABLE 3

Protein name (gene name)
1. alpha 1 type XXIV collagen precursor
2. alpha-fetoprotein enhancer binding protein
3. cell division cycle 2-like 5 (cholinesterase-related cell division controller) (CDC2L5)
4. chromodomain-helicase-DNA-binding protein 6 isoform X3 (CHD6)
5. chromosome 3 open reading frame 15 (C3orf15)
6. cilia- and flagella-associated protein 91 isoform 3
7. CLIP-associating protein 2 isoform X24 (CLASP2)
8. coiled-coil domain-containing protein 18 isoform X2 (CCDC18)
9. collagen triple helix repeat-containing RP11-45B20.2
10. collagen, type XXIV, alpha 1 (COL24A1)
11. death inducer-obliterators-3 (DIDO3)
12. diacylglycerol kinase iota isoform X4 (DGKI)
13. DNA methyltransferase 1-associated protein 1
14. DNAH10 variant protein
15. E3 ubiquitin-protein ligase MIB2 isoform X5 (MIB2)
16. growth-inhibiting protein 24 (GIG24)
17. hemicentin
18. histone acetyltransferase MORF alpha
19. IF2 protein (IF2)
20. immunoglobulin lambda light chain VLJ region (IGL)
21. kinesin-like protein (KIF2)
22. kinesin-like protein KIF21A isoform X7 (KIF21A)
23. laminin subunit alpha-2 isoform X3 (LAMA2)
24. LIM and calponin homology domains-containing protein 1 isoform X20 (LIMCH1)
25. LIM and calponin homology domains-containing protein 1 isoform X22 (LIMCH1)
26. lysine-specific demethylase 2B isoform X7 (KDM2B)
27. multiple endocrine neoplasia type 1 candidate protein number 18 (HSPF2)
28. MYCBP-associated protein isoform X1 (MYCBPAP)
29. nebulin-related anchoring protein, isoform CRA_b
30. Non-canonical poly(A) RNA polymerase PAPD5 (PAPD5)
31. PAP associated domain containing protein 5 variant
32. partitioning-defective 3-like protein splice variant c (PAR3L)
33. PF6
34. Pleckstrin homology-like domain family B member 2 (PHLDB2)
35. poly (ADP-ribose) glycohydrolase (PARG)
36. probable E3 ubiquitin-protein ligase HERC1 isoform X1 (HERC1)
37. probable E3 ubiquitin-protein ligase HERC1 isoform X11 (HERC1)
38. Protein AMBP (AMBP)
39. RUNDC1 protein (RUNDC1)
40. serine/threonine-protein phosphatase 4 regulatory subunit 3A isoform 1
41. serine-protein kinase ATM isoform X5 (ATM)
42. SHMT2 protein (SHMT2)
43. smoothelin-B3 (SMTN)
44. somatostatin receptor interacting protein splice variant a (SSTRIP)
45. sperm-associated antigen 17 isoform X3 (SPAG17)

TABLE 3-continued

Panel of protein biomarker covariates for algorithm that improve algorithm performance in FEV₁ predictions over a period 4 months or more after biomarker collection.

Protein name (gene name)

46. T-cell lymphoma invasion and metastasis 1 (TIAM1)
47. terminal nucleotidyltransferase 4B isoform b
48. tetratricopeptide repeat protein 21A isoform X6 (TTC21A)
49. thrombospondin 1 (THBS1)
50. ubiquitin carboxyl-terminal hydrolase 40 isoform X1 (USP40)
51. Voltage-dependent N-type calcium channel subunit alpha-1B (CACNA1B)
52. Zinc finger homeobox protein 4 (ZFHX4)

In one aspect, the method may comprise the step of determining an expression level of one or more, or two or more, or three or more, or four or more, or five or more, or six or more, or seven or more, or 8 or more, or 9 or more, or 10 or more, or 11 or more, or 12 or more, or 13 or more, or 14 or more, or 15 or more or 16 or more, or 17 or more, or 18 or more, or 19 or more, or 20 or more, or 21, or more, or 22 or more, or 23 or more, or 24 or more, or 25 or more, or 26 or more, or 27 or more, or 28 or more, or 29 or more, or 30 or more, or 31 or more, or 32 or more, or 33 or more, or 34 or more, or 35 or more, or 36 or more, or 37 or more, or 38 or more, or 39 or more, or 40 or more, or 41 or more, or 42 or more, or 43 or more, or 44 or more, or 45 or more, or 46 or more, or 47 or more, or 48 or more, or 49 or more, or 50 or more, or each protein (or corresponding expression of each protein) in the protein set, wherein said method is predictive of one or more clinical parameters in said individual. It is intended that detection of a protein or protein expression level may be carried out using any method known in the art or hereafter developed which allows a determination or estimation of the relative expression or amount of a given protein.

In one aspect, the method may comprise the step of comparing said expression level to a control value to obtain a combined score and/or a risk probability score. A principle component or similar analysis that combines all the data from more than one or all of the markers may be used as well to generate a score. The combined score may be used to assess strength of association between the expression level of one or more of the aforementioned proteins, and the clinical parameter of interest. In one aspect, the clinical parameter may be lung function decline. The risk probability score may be used to predict the degree of risk that an individual will have or develop lung function changes or other clinical events that are of interest during the progression of cystic fibrosis.

In one aspect, the one or more clinical parameters may be selected from FEV₁, BMI, PE, number of hospitalizations, antibiotic status, infection status, and/or other clinical feature of the disease. These parameters may be selected using statistical methods described herein.

In one aspect, the clinical parameter is lung function decline, wherein an individual classified as being high risk for rapid lung function decline is treated via more aggressive anti-inflammatory therapy and increased monitoring. It is well known that increased disease monitoring is associated with improved pulmonary status. Thus, those identified to be at risk for rapid decline would warrant more frequent clinical encounters to ensure that stability is maintained. In addition, those at risk of decline may have become colonized with new pathogenic bacteria. Identification of subjects at risk of pulmonary decline would trigger aggressive testing for new pathogens and treatment to stem pulmonary decline,

for example, by administration of anti-bacterial agents that address the specific pathogenic bacteria in an individual.

In one aspect, the sample may be blood, serum, urine, plasma, PBMCs, BALF, nasal and/or lower airway brushings, sputum, GI biopsies, lung explants, and combinations thereof. The sample may be obtained using routine methods known in the art. Multiple samples may be obtained over a period of time, for example, once every day, once every other day, once a week, once every two weeks, once every three weeks, once monthly, or once every two months, or once every three months, etc.

In one aspect, the detection step is carried out using mass spectrometry. For example, electrospray/matrix-assisted laser desorption ionization mass spectrometry may be used, as described herein.

In one aspect, the methods described herein may be carried out via the use of a computerized device. For example, one or more of the combined score or risk probability score may be calculated using a computer.

In one aspect, the combined score and risk probability score may be used to create a predictive model within a web browser, the computer having a graphical user interface (GUI) in which an end user can interactively explore the predictive model within a web browser. In one aspect, the end user may be, for example, a physician, patient, or patient guardian. The end user may use the interface to interactively explore the predictive model within a web browser. In one aspect, the GUI may be called the Cystic Fibrosis Point of Personalized Detection (CFPOPD). Predictions may be generated on an individual basis, utilizing data from CF cohorts, such as the Cystic Fibrosis Foundation Patient Registry, and the user may select which patient for which the prediction model is graphically illustrated. Inputs include, but are not limited to, clinical and demographic characteristics, such as those from electronic health records, and large-scale data from proteomics.

Measured and predicted lung function may be portrayed using three interactive graphs linked by a common timeline. FEV₁, the FEV₁ derivative, and the risk of rapid decline may all be displayed with corresponding shaded confidence bands. Additionally, the GUI may include patient-level and ecological descriptive variables as well as proteomic data that can be used to subset the pool of individual patients to select from, and the GUI may be expanded to include additional inputs. This may be used to facilitate, for example, the comparison of individually forecasted rapid lung function decline among individuals that are identical with respect to all model covariates expect for their sex. Both static and temporal covariates are displayed using graphical and text-based panels. An exemplary GUI prototype is currently available for public use at <http://cfpopd.amazon-shiny.duckdns.org/> and a screenshot is contained in FIG. 13.

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In certain aspects, the methods may be used in conjunction with evaluation of a drug or treatment for cystic fibrosis. For example, a potential treatment may be administered to an individual in need thereof, and the disclosed methods may be carried out following administration of such drug or treatment.

In one aspect, a method for predicting evolution of a clinical parameter in an individual diagnosed with cystic fibrosis is disclosed. In this aspect, the method may comprise the steps of calculating a risk probability score from expression levels of a biomarker set comprising one or more biomarkers of Table 1; and classifying said individual into a high risk or low risk group based on the risk probability score; forecasting lung function trajectory based on said risk probability score. In certain aspects, the step may be performed on a computer. In one aspect, the clinical parameter may be lung function decline. In one aspect, the combined score may be based on at least two clinical measurements, or at least three clinical measurements, or at least four clinical measurements. The biomarker expression may be detected using methods known in the art, or using methods as set forth herein.

In one aspect, a method for segregating severe and disease in an individual diagnosed with cystic fibrosis is disclosed. The method may comprise the steps of detecting and quantifying a biomarker set comprising one or more proteins of Table 1, wherein said detecting step is carried out by assaying a biological sample from said individual and a clinical parameter obtained from said individual; and classifying the individual into a high or low risk group based on the combined score. The steps may be performed on a computer. In one aspect, the clinical parameter may be forced expiratory volume (FEV₁), and the method may be used to predict the risk of a rapid decline in FEV₁ in a said individual.

Calculation of Combined Score and Risk Probability Scores

A marginal score may be used to classify an individual as a high risk or low risk individual for rapid lung function decline and the combined score may be used to perform feature selection wherein proteins are selected that correspond to said individual's risk of rapid lung function decline.

Marginal Score (Risk of Rapid Decline as High or Low). In the first stage, Y_{ij} may be a random variable representing lung function for patient i at time t_{ij} ; $i=1 \dots N$; $j=1 \dots n_i$. For illustration, time refers to age (in years), but may be defined using other measures, including but not limited to: time since first pulmonary function measurement, in years; time since first serum collection, in years. In applying the calculation, it may be assumed Y_{ij} is observed over one or more occasions, creating a random vector for each patient, represented as Y_i . In observation, these data may be sparsely or irregularly collected for various reasons, at the discretion of the user. Using Equation 1 (from Yao, Muller and Wang (2005)), the collection of lung function trajectories across patients observed at different time points can be expressed as follows. In one aspect, $X_i(T_{ij})$ may be the longitudinal process of FEV₁ for patient i at random time T_{ij} , which is measured with error ϵ_{ij} . This corresponds to the decomposition of longitudinal FEV₁ of Equation (1):

$$Y_{ij} = X_i(T_{ij}) + \epsilon_{ij} = \mu(T_{ij}) + \sum_{k=1}^{\infty} \xi_{ik} \phi_k(T_{ij}) + \epsilon_{ij}$$

where ϵ_{ij} is residual error with mean zero and some variance σ^2 ; $X_i(\bullet)$, which is the function depicting the

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smooth, continuous longitudinal FEV₁ profile for patient i , can be characterized, for example, using a cubic b-spline basis with knot locations as described by Szczesniak and others (2013). Applying the PACE approach (principal components analysis through conditional expectation) as described by Yao, Muller and Wang (2005) to the longitudinal FEV₁ data outlined herein can be decomposed into functional principal component scores $\xi_{i1}, \dots, \xi_{iK}$, where each score is a univariate quantity representing the k^{th} harmonic depicting the i^{th} patient's continuous longitudinal FEV₁ profile. These quantities correspond to said marginal scores and have been calculated previously for FEV₁ data from the Cystic Fibrosis Foundation Patient Registry, and the resulting scores can be used to classify said individual as high risk or low risk for rapid decline (Szczesniak et al. 2017). Techniques for choosing the number of harmonics, K , include, but are not limited to: cross validation.

Combined Score (Feature Selection According to Expression Level of Protein and Risk of Rapid Decline). In one aspect, the 1^{st} protein of the k^{th} patient may be represented as P_{ik} , and P_i may be the vector combining measurements of this protein across all patients. Similarly, the functional principal component scores (said marginal scores) for the k^{th} component across all patients may be the vector $\xi_{k \cdot pkl}$ may be the correlation coefficient representing the bivariate association between ξ_{ik} and P_{ik} . Methods to estimate ρ_{kl} include, but are not limited to, Spearman's rank correlation coefficient and Pearson's correlation coefficient, such application being readily understood by one of ordinary skill in the art. Said coefficient ρ_{kl} , with estimated value denoted as $\hat{\rho}_{kl}$, represents said combined score from this two-stage process for a given protein with the FEV₁ trajectory. Implementation of the approach can proceed using R (R Foundation for Statistical Computing, Vienna, Austria).

In one aspect, a risk probability score may be obtained using the following Equation (2):

$$Y_{ij} = X_i(T_{ij}, Z_{ij}) + \epsilon_{ij} = \mu(T_{ij}, Z_{ij}) + \sum_{k=1}^{\infty} \xi_{ik} \phi_k(T_{ij}, Z_{ij}) + \epsilon_{ij},$$

Obtaining said combined score may further include inputting data for one or more clinical parameters of the individual. The marginal and combined scores, further including the step of inputting data on one or more clinical parameters, can be derived from a two-stage process. Stage I. Let variables be defined as in Stage I. Additionally, let input data from one or more clinical parameters, such as observed body mass index percentile, be represented as Z_{ij} . The input will take the form of a vector if inputting data from one clinical parameter, and will take the form of a matrix with dimension $N \times C$ if there are $c=1, \dots, C$ clinical parameters. One or more covariates may be time-varying. Using technique and assumptions for mean-adjusted functional principal components analysis described by Jang and Wang, Equation (1) is expanded as Equation (2) above, where estimation can proceed as they describe based on Equation (2) above with stated changes to covariance accommodating Z_{it} . Let ξ_{ikc} be the functional principal component scores, where c denotes that these scores are covariate adjusted. Stage II. As described in Stage II, ρ_{kl} is estimated for bivariate association of resulting covariate-adjusted functional principal component scores, in which bivariate association is estimated between vectors ξ_{kc} and P_i . Resulting estimate $\hat{\rho}_{kcl}$

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represents combined score for a given protein with the FEV₁ trajectory, which further includes inputting data on one or more clinical parameters.

Obtaining a risk probability score may further include the step of inputting data for one or more clinical parameters of an individual.

Modifying notation from Stage I, Y_i may be realizations of a longitudinal patient-specific process other than FEV₁, such as body mass index percentile. By Equation (1) and aforementioned approaches, this substitution yields principal component scores $\xi_{i1o}, \dots, \xi_{iKo}$, where o indicates the given outcome process being analyzed as Y_i . Stage II. Modifying notation from Stage II, ρ_{klo} may be the correlation coefficient representing the bivariate association between ξ_{ko} and P_k . Using the method described in Stage II, coefficient ρ_{klo} , with estimated value denoted as $\hat{\rho}_{klo}$, represents said combined score from this two-stage process for a given protein with the outcome process noted as o .

Additional clinical parameters as inputs. If using additional clinical inputs as described above and letting $Y_i(t)$ be realizations of a longitudinal patient-specific process over time t other than FEV₁, such as body mass index percentile, yields covariate-adjusted combined scores for a different clinical endpoint. Resulting estimate $\hat{\rho}_{klo}$ represents combined score for a given protein with the outcome process noted as o , which further includes inputting data on one or more clinical parameters as covariates.

Derivation of Risk Probability Score. Let Y_{ij} represent FEV₁ for a given patient and time point. Assume that the longitudinal FEV₁ process follows the Gaussian linear mixed model with non-stationary covariance as defined by Diggle, Sousa and Asar (2015). The prediction algorithm to obtain a risk probability score can be defined through the following sequence of equations:

$$Y_{ij} = f_i(t_{ij}) + U_i + W_i(t_{ij}) + \epsilon_{ij} = g(t_{ij}) + P_{ij}^T \beta + U_i + W_i(t_{ij}) + \epsilon_{ij}, \quad \text{Equation (3):}$$

where $f_i(t_{ij})$ is the mean response function for the patient's longitudinal lung function process; U_i is a patient-specific random intercept term, allowing patient profiles to deviate randomly from one another; $W_i(t_{ij})$ is a stochastic process characterizing the change in an individual's lung function process that cannot be explained by $f_i(t_{ij})$ alone; ϵ_{ij} is residual error from the model. In the second part of the equation, $g(t_{ij})$ is a nonparametric function representing a smooth, continuous-time process for lung function, which may be estimated using cubic b-splines; P_{ij} is the covariate information on a single protein or set of proteins, corresponding to a vector or matrix, where T implies taking the transpose; β is the vector of parameter coefficients corresponding to associations between lung function and protein expression level(s). Assuming $W_i(t_{ij})$ follows integrated Brownian motion, it follows that the derivative of this quantity in Equation (3) will yield Brownian motion, denoted $B_i(t_{ij})$ for the i^{th} subject and j^{th} time point. Then, the risk probability score is defined as Equation (4):

$$P_r(B_i(t_{ij})) = P_r(B_i(t_{ij}) < \delta_i - f_i'(t) \mathcal{H}_{iP}(t))$$

where δ_i is certain threshold for the rate of decline, which may be patient specific (hence i subscript) or uniform for all patients; $f_i'(t)$ is the first derivative of the process from the model structure in Equation (3) and $\mathcal{H}_i(t)$ is all protein expression covariate history before a given time on the i^{th} patient. Protein expression levels may either be observed cross-sectionally or longitudinally.

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Quantifying Uncertainty of Risk Probability Score. Pointwise confidence intervals for the patient-specific risk probability score defined in Equation (5) can be derived as follows. Without loss of generality, assume this derivation is for a 95% pointwise confidence interval. Let μ_{B_i} and Σ_{n_i} denote the mean and variance for $B_i(t_{ij})$ given $\mathcal{H}_i(t)$. We assume the mean μ_{B_i} satisfies:

$$\sqrt{n_i}(\hat{\mu}_{B_i} - \mu_{B_i}) \xrightarrow{D} N_{n_i}(0, \Sigma_{n_i}) \quad \text{Equation (5):}$$

where μ_{B_i} denotes the vector $\mu_{B_{i1}}, \dots, \mu_{B_{in_i}}$, $\hat{\mu}_{B_i}$ is the estimate of μ_{B_i} , and n_i is the sample size for i^{th} patient.

For each patient i , let $\hat{\Sigma}_{n_i}$ denote the estimation of Σ_{n_i} from estimating parameters in the model from Equation (3) and taking derivatives of the relevant probability risk score quantities in Equation (4). The bootstrapping process to obtain a 95% confidence interval for risk probability is below.

(1) Sample B—Independent samples, $Y_{n_{i1}}, \dots, Y_{n_{iB}}$ from $N_{n_i}(0, \Sigma_{n_i})$, $B=1, \dots, 100$, and define parameter $\mu_{B_i}^* = \mu_{B_i} + n_i^{-1/2} Y_{n_{iB}}$.

(2) For each sample $Y_{n_{iB}}$, calculate $Pr^*(B_i(t_{ij}))$ using $\mu_{B_i}^*$ by Step (1) above and calculate mean square error (MSE) for

$$Pr^*(B_i(t_{ij})) \text{ as } MSE_{n_i} = B^{-1} \sum_{b=1}^B [Pr^*(B_i(t_{ij})) - Pr(B_i(t_{ij}))]^2.$$

(3) Construct the 95% confidence interval for risk probability by $P_r(\hat{B}_i(t_{ij})) \pm z_{0.975} \sqrt{MSE_{n_i}}$.

In one aspect, the risk probability score may be calculated using an algorithm which further includes the step of inputting data from one or more clinical parameters of an individual.

Derivation of Risk Probability Score. Additionally define Z_{ij} , wherein additional inputting data on clinical parameters, such as observed body mass index percentile, are represented as Z_{ij} . The input will take the form of a vector if inputting data from one clinical parameter, and will take the form of a matrix with dimension $N \times C$ if there are $c=1, \dots, C$ clinical parameters, and Z_{ij} may include covariates that are time-varying. To accommodate the additional inputs, Equation (3) can be expanded as:

$$Y_{ij} = f_i(t_{ij}) + U_i + W_i(t_{ij}) + \epsilon_{ij} = g(t_{ij}) + P_{ij}^T \beta + Z_{ij}^T \beta_Z + U_i + W_i(t_{ij}) + \epsilon_{ij} \quad \text{Equation (6):}$$

where β_Z is the parameter vector representing associations between each clinical input variable and the longitudinal lung function process.

It follows that the risk probability score can be modified to include covariate history by expanding Equation (4) as follows:

$$P_r(B_i(t_{ij})) = P_r(B_i(t_{ij}) < \delta_i - f_i'(t) \mathcal{H}_{iP}(t), \mathcal{H}_{iZ}(t)) \quad \text{Equation (7):}$$

wherein covariate history from the clinical, protein expression, and previous outcome inputs are represented as $\mathcal{H}_{iP}(t)$ and $\mathcal{H}_{iZ}(t)$, respectively. Although it is suppressed in Equation (7), the expression includes previous outcome history prior to the time point t_{ij} .

Quantifying Uncertainty of Risk Probability Score. Bootstrapped estimates for covariate-adjusted risk probability scores can be computed as in Equation (5). In one aspect, the risk probability score may be performed for another clinical outcome.

Modifying notation from above, let Y_{ij} be realizations of a longitudinal patient-specific process other than FEV₁, such

as body mass index percentile. By Equations (3) and (4) and approaches therein, the patient-specific risk probability score can be expressed as:

$$P_r(B_{io}(t_{ij})) = P_r(B_{io}(t_{ij}) < \delta_{io} - f'_{io}(t) | \mathcal{H}_{io}(t)) \quad \text{Equation (8):}$$

where elements correspond to the terms defined in Equation (4) but for a different outcome o than FEV_1 . Boot-strapped estimates for risk probability scores based on outcome o can be computed as in Equation (6).

Equation (8) and approaches herein can be adapted using the procedure set forth above to acquire covariate-adjusted estimates of risk probability for clinical outcome o .

In one aspect, disclosed herein is a method for treating an individual at risk for non-linear lung function decline. The method may comprise a) determining one or more covariates associated with lung function in said individual, said covariate being selected from one or more of a clinical measure, a biomarker or an imaging marker; b) calculating a risk probability score based on said determining of one or more covariate, said risk probability score being used to characterize an individual as having no predicted lung impairment, mild predicted lung impairment, moderate predicted lung impairment, or severe predicted lung impairment; and c) treating said individual characterized as having mild predicted lung impairment, moderate predicted lung impairment, or severe predicted lung impairment with one or more of increased frequency of disease monitoring, increased frequency of infection monitoring, anti-inflammatory therapy, or combinations thereof.

In one aspect, the risk probability score may comprise a risk probability of a clinical outcome selected from a decrease in forced expiratory volume in one second (FEV_1), pulmonary exacerbation (PE) frequency, FEV1-indicated exacerbation signal (FIES), ventilation defect percent (VDP), abnormal lung heterogeneity in lung as measured via imaging, lung hyperinflation, and combinations thereof.

In one aspect, the covariate may be a clinical measure selected from forced expiratory volume in one second (FEV_1), body mass index percentile (BMI), pulmonary exacerbation (PE) frequency, historic lung function, ventilation defect percent (VDP), partial ventilation data, ventilation heterogeneity data, hyper-intensity data, FEV1-indicated exacerbation signal (FIES), and combinations thereof.

In one aspect, the covariate may be a biomarker described in Table 1, Table 2, Table 3, or Table 4, said determining comprising determining an expression level of a protein described in Table 1, Table 2, Table 3, or Table 4.

In one aspect, the covariate may be an imaging marker, said imaging marker including but not limited to VDP, lung heterogeneity, lung hyperinflation, CT, or the like.

In one aspect, the imaging marker may be a functional lung measurement, a structural lung measurement, or combinations thereof.

In one aspect, the imaging marker may be obtained by an imaging method.

In one aspect, the image technique may be selected from hyperpolarized (HP) ^{129}Xe , Ultra-short Echo-time (UTE) Magnetic resonance imaging (MRI), computed tomography (CT), and combinations thereof.

In one aspect, the imaging marker may be structural remodeling of the lung.

In one aspect, the imaging marker may be bronchiectasis.

In one aspect, the predicted lung impairment may be non-linear decline in one or more parameters selected from

Ventilation Defect Percentage (VDP), FEV_1 , partial ventilation, ventilation heterogeneity, ventilation hyper-intensity or a combination thereof.

In one aspect, the individual may have normal FEV_1 at the time of said determining and said calculating.

In one aspect, the risk probability score may predict the probability of lung function decline over a period selected from three weeks to four months, or six months, or twelve months.

In one aspect, rapid (non-linear) lung function decline may be defined as a rate of change in longitudinal FEV_1 that falls below 1.5% predicted/year.

In one aspect, non-linear lung function decline may be defined by the FEV1-indicated exacerbation signal (FIES) score.

In one aspect, the one or more covariates may comprise a time-varying covariate.

In one aspect, the time-varying covariate may comprise infections with Pa, MRSA, CF-related diabetes and use of state insurance as a marker of socioeconomic status.

In one aspect, the biomarker may be one or more biomarkers selected from Table 1.

In one aspect, the biomarker may be one or more biomarkers selected from Table 2.

In one aspect, the biomarker may be one or more biomarkers selected from Table 3.

In one aspect, the biomarker may be one or more biomarkers selected from Table 4.

In one aspect, the individual may be diagnosed with cystic fibrosis (CF).

In one aspect, the individual may be diagnosed with cystic fibrosis (CF) and has a normal forced expiratory volume in one second (FEV_1) as measured by spirometry.

In one aspect, the individual may be a pediatric patient having cystic fibrosis (CF).

In one aspect, the individual may be a patient under the age of 13 years of age and having cystic fibrosis (CF).

In one aspect, the method may forecast lung function trajectory for a period of about three weeks to about six months, or about six months to about twelve months.

In one aspect, the method may further comprise applying a statistical algorithm to estimate correlation between a covariate value and predicted lung function.

In one aspect, the method may be carried out via a computer system, and wherein said method comprises capturing and displaying information related to said characterization of said individual using a graphical user interface (GUI).

In one aspect, the method may further comprise assessing a variable selected from one or more of sex, body mass index (BMI), pulmonary exacerbation (PE), number of hospitalizations, antibiotic status, infection status, and combinations thereof, in said individual.

In one aspect, the lung function decline may be defined by one or more of absolute change in FEV_1 , rate of decline, risk of rapid decline, FIES.

EXAMPLES

The following non-limiting examples are provided to further illustrate embodiments of the invention disclosed herein. It should be appreciated by those of skill in the art that the techniques disclosed in the examples that follow represent approaches that have been found to function well in the practice of the invention, and thus can be considered to constitute examples of modes for its practice. However, those of skill in the art should, in light of the present

disclosure, appreciate that many changes can be made in the specific embodiments that are disclosed and still obtain a like or similar result without departing from the spirit and scope of the invention.

Applicant analyzed serum samples from CF patients with severe (n=44) or mild (n=44) lung disease. Serum protein was purified via gel separation and liquid chromatography. Protein isoforms were identified using a cut off of at least 100 mass signatures per protein, coverage ranging from between about 15 and 100%. A combination of gel-based and MS-based label free quantitation was used to quantify the proteins. Pathways including the dataset members were analyzed.

In an exhaustive analysis of 61,942 serum protein isoforms, Applicant discovered a number of novel markers of CF lung disease severity and inflammation and found that combining their measurement (either grouped or individually depending on the marker) with Functional Data (FD) Analysis of FEV₁ can predict lung function decline by 6-18 months in advance. Applicant hypothesized that characterization of blood protein expression and modification in longitudinal samples integrated with monitoring of FEV₁ changes over time and through FD analysis can provide a sensitive and specific algorithm that predicts risk of CF lung disease progression which can be used to inform therapeutic intervention. Applicant has developed a novel statistical model to predict FEV₁ decline that have now been modified to include the novel biomarkers, and which can be adapted to include use of other clinical parameters besides FEV₁ to predict disease progression.

In particular, Applicant has identified and validated serum proteome changes in banked samples collected from patients with stable and declining FEV₁, developed a dynamic prediction model by integrating analyses of validated proteomic data with Functional Data (FD) analysis of longitudinal FEV₁ data to produce a novel diagnostic algorithm that identifies individuals at risk of lung function decline, and has evaluated the capacity of dynamic prediction modeling to identify CF patients who develop rapid pulmonary decline during adolescence in banked patient samples, testing the performance of the markers in banked samples from the EPIC study. Applicant's studies have the potential to fundamentally shift the nature of CF care from an evidence-based model of care decisions to individual patient-based care decisions informed through a predictive molecular and lung function disease platform and/or other clinical parameters. The disclosed algorithms of predictive biomarkers may be used to inform therapy choices and significantly improve CF care.

Proteomic studies may be conducted with LC MS/MS tandem mass spectrometric examination of patient serum. Each sample may be split into 3 fractions which can be analyzed by MS for 15 hours (including 3 technical replicates, 3 quality controls, and 2 wash runs preventing carry over). This allows deep screens to be conducted that capture information on approximately 7,000-10,000 protein isoforms and modification per sample with high confidence and accuracy and allows capture of low-level proteins where differences are more often found when comparing disease severity cohorts. These analyses are far more rigorous than the usual "service center" analysis, generating approximately 7-10 times more data. This increased data collection also increases the time for analysis (~10 fold) but provides rigor for detection of useful biomarkers and decreases false discovery rates, markedly increasing chances of success. For Functional Data (FD) and Functional Principle Component (FPC) analyses of FEV₁, Applicant's preliminary analyses

build upon FD analysis and longitudinal models that have been applied to the CF Foundation Patient Registry (CFF-PR). Further analyses of the available cohort data at hand, providing strong evidence of associations and feasibility. The modeling disclosed herein blends established biostatistical approaches with modern FD approaches to characterize the nonlinear FEV₁ trajectory of individual patients and predicts subsequent decline. The disclosed methods may be used for in-clinic applications for decision aids for pre-clinic planning and at the time of patient encounters. Marker data correlation with FEV₁ was measured in multiple simulations and by appropriate statistical tests as described.

As noted above, three measures are generally used in monitoring CF disease progression and response to therapy: FEV₁, PE frequency, and BML. Although these measures (particularly) lung function (FEV₁), have demonstrated steady improvement over the past two decades, a rapid deterioration of lung function persists, especially during adolescence (FIG. 3). See also https://cff.org/Research/Researcher-Resources/Patient-Registry/2019-Patient-Registry-Annual-Data_Report.pdf. While there are several potential interpretations for this observation, without intending to be limited by theory, it is believed that systemic changes reflected in the blood culminate in measurable pulmonary decline during this period (10-19 years of age).

Care for CF patients has advanced rapidly over the past two decades, with an unprecedented increase in patient longevity and quality of life. The reasons for these improvements include: a robust CFF-PR that collects data from nearly all CF patients in the US, allowing assessment of outcomes and treatment responses; standardization of care & therapies; and dramatic advances in new CF therapeutics, especially genotype-specific CFTR modulators such as KALYDECO® (4) and, to a lesser extent, ORKAMBI® (5), and more recently TRIKAFTA® (6). These advancements have increased the median survival of CF patients to 41 years (CFF-PR—2014), and >50% of CF patients alive today are adults. In many respects, research and care to improve CF patient outcomes has become the model for other rare diseases.

Applicant modified proteomic approaches (7-17) to examine serum from CF patients with mild or severe lung disease (13). Based on the experience of others in the cancer and cardiovascular fields, where conventional "shotgun" proteomics (which identifies 500-1000 proteins per sample) failed to distinguish disease severity in cohorts exhibiting the same disease, Applicant hypothesized that deeper screens would be necessary for CF studies. A proteomic analysis protocol was developed, involving multidimensional separation of proteins by abundant protein-adsorption columns and gel and column chromatography. This approach generated multiple fractions from each sample that were subjected to 15 hours of mass spectrometric analysis.

The following is a description of an exemplary proteomic analysis of CF patient serum. The approach may be applied to the analysis of any biological sample, for example, plasma, urine, cells, and other tissues. The methodology involves novel modifications of previously described approaches (Ziady A G, Sokolow A, Shank S, Corey D, Myers R, Plafker S, Kelley T J. Interaction with CREB binding protein modulates the activities of Nrf2 and NF-kappaB in cystic fibrosis airway epithelial cells. *Am.J.Physiol Lung Cell Mol.Physiol* 2012 June 1; 302(11): L1221-L1231. PMID:PMC3379036; Chen J, Kinter M, Shank S, Cotton C, Kelley T J, Ziady A G. Dysfunction of Nrf-2 in CF epithelia leads to excess intracellular H₂O₂ and inflammatory cytokine production. *PLoS.One*. 2008; 3(10):

e3367; Ziady A G, Kinter M. Protein sequencing with tandem mass spectrometry. *Methods Mol.Biol.* 2009; 544: 325-41). Biological fluid or tissue lysate can be adsorbed for albumin or other highly expressed proteins using absorption columns. Protein may be precipitated from eluate or lysate with acetone (equilibrated to 90% acetone), dried, and rehydrated at a concentration of 5 mg/ml in 8 M urea, 2% CHAPS, 50 mM DTT in water. Greater than 45 µg of protein for a sample is generally subjected to SDS-PAGE, excised gel fragments containing all the protein from the sample are reduced and alkylated (to achieve more fractionation more gel fragments are cut), and subjected to in-gel tryptic digestion (20 µg/ml), and protein peptides are extracted for LC MS-MS. Extracts were acidified by equilibration in 0.1%-1% acid (e.g. acetic or formic acids), loaded onto a fractionation column (e.g. Thermo Fisher Scientific Acclaim PepMap C₁₈ column) at a flowrate of 0.15-0.5 µl/min, and subjected to nanospray tandem mass spectrometry with a mass spectrometer (e.g. Thermo Fisher Scientific LTQ Velos Pro spectrometer). Analysis is conducted in data-dependent mode capturing the 3-12 most abundant parent ions from full MS scans for fragmentation by collision induced dissociation (CID). Each gel fragment is run 3-9 times, and the data files from all runs are pooled for database analyses. In addition to multiple fractionation steps mass tolerance of 0.1-2.0 Da is used for parent ions and 0.01-0.7 Da for fragment ions to enhance the ability to quantitate low level proteins. Identified protein isoforms using 2-100% coverage cut off with >2 mass signatures per protein.

The EPIC Observational study includes >1,000 patients with annual blood samples for >10 years linked to the CFF-PR. 44 mild (FEV₁>85th percentile among CF patients) and 44 severe (<45th percentile) patient cohorts were matched based on age, gender, genotype, and *P. aeruginosa* infection status, then randomized and blinded before proteomic analysis. Applicant identified a total of 61,942 protein isoforms expressed across both cohorts, with 4751 proteins identified in at least half the subjects of each cohort. Data were normalized to a relative abundance (RA) measure (0 to 1) for each sample. For each isoform, RA was summarized (mean RA, number with RA >0, ratio of RA mild/severe) and, to reduce the data, a battery of paired statistical tests was performed on the matched samples: McNemar's, Wilcoxon Signed-Rank, paired Student's *t* test, and permutation of the difference.

In addition, a two-fold approach was used to identify protein isoforms as candidates for improved prediction of FEV₁. Functional principal components (FPC) scores were correlated with each protein expression level as described, measured using correlation coefficient $\hat{\rho}_{kct}$. In addition, these isoforms were included as covariates in one-at-a-time modeling of Equation (3). These analyses generated the newly discovered 44 isoforms above which are the first serum biomarkers of CF disease severity identified by nonbiased analysis and are the basis for developing a lung function decline prediction model. Furthermore, the protein isoforms identified are baseline biomarkers of disease that segregate with disease severity when patient are stable. This is far more advantageous than presently available biomarkers such as C-reactive protein, which are only useful during acute exacerbation (20-24). The initial twenty biomarkers identified exhibited areas under the curve (AUC) of 0.69 or higher. These AUC values indicate that the biomarkers are superior to presently available markers of disease (21-24). Many of the biomarkers have either physiological or biological connections to CF (FIG. 4). For example, tight junction protein (TJP) 3 contains a PDZ motif and is a

member of a family of proteins that interact with CFTR (25). Increases in serum IgG Kappa light chain levels are associated with chronic inflammation (26). Alpha-1-acid glycoprotein 2 is one of 4 biomarkers used to predict inflammation and mortality beyond CF (27), and transthyretin has been shown to be significantly decreased in CF patients compared to non-CF (28). Excessive inflammatory signaling in CF and an increase in patients with severe disease would be expected and is suggested by nonbiased biomarker identification.

CF lung disease demonstrates a steady decline that typically manifests during adolescence and young adulthood (29); however, rapid decline, characterized by accelerated loss of lung function relative to center-specific and/or population norms, is a ubiquitous event in the lives of patients (FIG. 1). Identifying when patients are at highest risk for rapid lung function decline is a significant gap in CF research, and offers the opportunity to intervene prior to irreversible lung damage. A key contributor to this gap has been the paucity of individualized predictive data on the specific timing and severity of lung function decline, and this is sustained by continued reliance on linear statistical approaches to fit nonlinear lung function decline (30). Moreover, phenotypes of patients at risk for rapid decline have not been well defined due to the analytically complex progression of CF (31).

In contrast to historical approaches, Applicant's biostatistical research on CF lung disease progression has been based on nonlinear longitudinal data analysis methodology and implementation in CF clinical/translational research. Applicant's approach fuses longitudinal data analysis together with FD analysis (32), a branch of statistics that offers tools to characterize nonlinear phenomena. FD analysis sheds light on complex pathophysiological relationships in different disease states, such as sleep disorders (33) and Alzheimer's disease (34).

Applicant has used the CFF-PR to demonstrate that FD analysis predicts CF lung-function decline with improved accuracy compared to traditional linear approaches (35) and can be used to identify phenotypes of rapid decline. Historically, CF registries (such as the CFF-PR) have carefully maintained lung function data to track disease, but have not utilized data for prognostic care (36). The disclosed dynamic prediction models using FD analysis have leveraged this rich longitudinal data to develop more accurate tools that predict disease course and in turn help prioritize interventions for the individual patient (FIG. 5). The disclosed methods meaningfully contribute to CF precision medicine, an area that has been noticeably understudied (37).

Major advancements in personalized medicine in CF may change the clinical course of the disease (38). Applicant has successfully used FD analysis to characterize nonlinear population-level lung function decline of cystic fibrosis patients in the US. Applicant's CFF-PR study of over 30,000 patients and 500,000 FEV₁% measurements utilized longitudinal FD methods to address nonlinearity and serial correlation (35). Applicant estimated degree (velocity) and timing of rapid decline by taking derivatives in Applicant's model and traditional models. The covariance structure included exponential decay and random intercepts. The FD model was superior, compared to traditional models, in terms of estimating the onset and severity of rapid FEV₁% decline and model validity (FIG. 5). Applicant's results indicate that nonlinear, heterogeneous lung disease progression prevails, and the ability of FD models to establish CF phenotypes predictive of rapid decline.

Applicant has further identified FEV₁ phenotypes corresponding to early, middle and late rapid decline in patients 6-21 years of age. Using the CFF-PR data, modes of variation in FEV₁ progression are characterized as functional principal components (39). The majority of variation (first functional principal component: 94%) among patient profiles are characterized by differences in mean longitudinal FEV₁ trajectories. Average degree of rapid decline was similar among phenotypes (roughly -3% predicted/year); however, average timing differed, with early, middle and late phenotypes experiencing rapid decline at 12.9, 16.3 and 18.5 years of age, respectively. Individuals with the late phenotype had the highest initial FEV₁ but experienced the greatest loss of lung function.

FEV₁ variation coupled with nonlinear progression over age produces an uneven, "saw-tooth" shape for each individual's trajectory (see FIG. 1), making it difficult to model the time-course of a patient's underlying lung function and utilize it for earlier detection and intervention. To further improve the predictive accuracy of rapid decline, Applicant utilized FD analysis to develop a dynamic prediction model of CF progression that accounts for the variation in FEV₁. To leverage existing data for individual patient prediction, a dynamic prediction model recently proposed by Diggle et al (40) was expanded to accommodate the jagged, nonlinear shape of FEV₁ trajectories using CFF-PR data on 27,296 patients >6 years who each had 1 to 89 FEV₁ measurements. Covariates in the model were birth cohort, copies of delf508, baseline FEV₁, and male gender. Time-varying covariates included infections with Pa, MRSA, CF-related diabetes and use of state insurance as a marker of socioeconomic status. Interactions between each of these covariates with time were examined. To account for irregular follow up and occurrence of PEs, rolling, time-varying covariates were included for the numbers of follow up visits and PEs within the last year. Although these covariates are not commonly used in the literature, their inclusion improved predictive performance of the model based on residual analyses. All covariates were statistically significant (P<0.05) with the exception of state insurance (P=0.07).

Applicant evaluated predictive ability of the proposed stochastic model using a preliminary sample size of 36 subjects with available data on proteomic markers. These subjects contributed a total of 1975 FEV₁ longitudinal observations. The subjects were randomly split, with 80% contributing measurements to the training dataset for model building, and 20% providing data for the validation step. Using the training dataset, a model that included terms to model nonlinear progression of FEV₁ over age (cubic b-splines), a severity indicator established by the EPIC study (binary variable) was used; also included were terms for the markers and their interaction with age. Severe classification met statistical significance (P<0.05).

Predictive accuracy, measured using Akaike information criterion (AIC), was superior in this model compared to a model that excluded the proteomic marker terms. Validation metrics that were assessed in the test cohort included mean absolute deviation (MAD), root mean-square error (RMSE), mean absolute percentage error (MAPE) and correlation between predicted values and observed values. Based on the FEV₁ scale, validation metrics showed relatively small prediction error. MAPE, which measures forecast accuracy as percent difference between actual FEV₁ and predicted FEV₁, shows that there is relatively small error between projected and actual FEV₁ values. Correlation between observed and

predicted values is excellent (above 0.80) and is significantly higher than presently available measures (range from 0.54-0.71).

Applicant also assessed the predictive value of proteomic markers using FD and dynamic prediction as described above. A number of the biomarkers were included as covariates and their interaction with age in the Reduced Model. Other covariates included in the model were age-specific components to fit the FEV₁ trajectory, as shown in the FD analysis in previous studies, and the Mild/Severe designation from Applicant's original study. The model showed that proteomic markers were significantly associated with mean FEV₁ (coefficient: -8.03, SE: 4.07, z=-2.0, P=0.04) and approached significance in their association with FEV₁ decline (coefficient for interaction: 1.03, SE: 0.79, z=1.3, P=0.19).

Serum proteome changes in banked samples collected from patients with stable and declining FEV₁ can be validated as follows. Preliminary studies generated serum biomarkers that segregate CF disease severity during periods of disease stability. Conduct of longitudinal analyses to validate the predictive power of biomarkers in samples from the EPIC CF cohort may provide the confidence for future modeling. Proteomic discovery studies in 88 cross sectional samples identified 20 biomarker changes that segregated CF patient lung disease severity in adolescence. Data collected from proteomic screens can benefit statistical models of lung function decline. To determine the utility of the molecular markers, top 20 molecular biomarkers were examined for their ability to enhance lung function decline modeling. Initial studies are cross sectional and conducted in discovery mode. Multiple (>10) simulations of functional principle component (FPCA) analyses of FEV₁ including data for each of the top 20 markers showed significant correlations between the behavior of novel markers and FEV₁ decline (FIG. 6).

The discovery cross sectional data was collected by high throughput discovery mode MS analysis. Data from both targeted MS and ELISA may be integrated with FD analysis, either in combination or separately. Biomarkers discovered in preliminary cross-sectional studies for prediction of future FEV₁ decline, may be validated using targeted proteomics and ELISA analyses of banked serum samples from the EPIC Observational study.

Samples. Analysis of samples from EPIC can be used to evaluate the predictive capacity of the biomarkers 1-5 years in advance of cohort segregation by FEV₁. These can be compared with additional candidate biomarkers, and secondary analyses can be used to examine relationships to established predictors of disease severity, including PEs, BMI, and microbiology. The choice of the EPIC samples may be based on the availability of longitudinal samples for patients that cover the age range where significant lung-function decline is observed (e.g., early teen years). The longitudinal proteomic profiles can be categorized for subjects already profiled at 'baseline' or time zero. This will determine if the baseline protein markers are able to predict subsequent disease progression. Preliminary data using FEV₁ alone shows a fairly even distribution of different phenotypes to examine: stable high (n=16) and stable low (n=13) (total stable, n=29) and rapid lung-function decline among 'mild' lung disease subjects at baseline (n=10) and in those with more severe disease at baseline (n=12) (total 'decliners', n=22). The remaining patients (29 of the original 88) may be used to supplement the 'extreme phenotype' data and provide information on proteomic profiles across the continuum of clinical presentation.

Mass spectrometry. Applicant has developed useful mass spectrometry-based approaches for the identification/quantitation of thousands of proteins in serum samples. Protein is prepared from serum using albumin adsorption columns and gel and column chromatography, followed by tryptic digestion. Following preparation of whole-serum protein, serum-protein peptides are extracted and subjected to data-dependent sequencing and mass spectrometry analysis, as previously described (33-33; 35; 37-41; 42). Briefly, the samples are loaded in a HPLC system autosampler and eluted by reverse-phase chromatography into a mass spectrometer fitted with a nanospray ion source for analysis. The mass analyzer is set up for a data-dependent mode using dynamic exclusion settings: repeat count=1; repeat duration=0.5 minutes; exclusion list size=50; exclusion duration=1.5 minutes; exclusion mass width=1.5 amu. Collision-induced dissociation (CID) is used to fragment peptides, and CID spectra are searched against a human fasta database using Proteome Discoverer™ software. A decoy database is used to control for false discovery. A threshold filter of >2.0 for peptide XCorr score is used for sequence identification. For preliminary studies, this produced a range of coverage for identified proteins of 5.15%-79.91% and an average of 10.03% for all proteins identified. This high stringency filter of data provides more reliable quantitation, and aids the reduction of data in follow-up statistical analyses.

Given the emphasis on early detection of rapid disease progression from the CF community, the availability of extensive demographic, clinical, molecular and environmental measurements from the EPIC data augmenting the CFF-PR, and the emergence of promising methodologic approaches for analysis, the disclosed methods allow for improved dynamic prediction models of rapid decline through integration of validated proteomic markers. Rosenfeld and colleagues (47) have utilized part of the EPIC cohort (n=946 patients who were Pa-negative at enrollment) to study associations between data collected in the year after the first pulmonary function test (PFT) and subsequent rate of FEV₁ decline (mean±SD follow up: 6.2±1.3 years). Through multivariable linear modeling of age-related FEV₁ progression with generalized estimating equations, they confirmed established risk factors for decline (e.g., female gender) and identified a new risk factor (*S. maltophilia*). Although this was a relatively young cohort (mean±SD age at entry: 7.9±2.0 years) with mild CF disease, they observed a similar “ceiling” effect to what was originally described by Konstan and colleagues and previously found by Applicant.

Modes of variation in FEV₁ decline for the 88 EPIC subjects (44 Mild, 44 Severe, described above) with longitudinal PFT and cross-sectional proteomic data were examined using functional principal components, the FD analysis technique to the CFF-PR, in order to characterize rapid decline during adolescence and adulthood. Median (IQR) age at entry was 12.5 (9.8-15.3) years; follow-up ranged from 20 to 216 PFTs per patient. Individual smooth curves from the FD analysis had substantial variation between patients and within an individual patient over time (FIG. 7). The first functional principal component explained 85.3% of the variation in FEV₁, suggesting that the individual trajectories were similar to the overall mean curve but shifted according to initial FEV₁. The four subsequently ordered functional principal components, which reflect higher-order modes of oscillatory variation, explained 8.1%, 3.7%, 1.8% and 1.0%, respectively.

Rapid decline was observed in different periods of age for the majority of individuals in the cohort. Most rapid decline during the observation period occurred, on average, at

15.5±2.3 years of age, with decline of 3.9±4.2% predicted/year. To understand how rapid decline in this cohort is related to proteomic markers, the association between each FD parameter and marker of interest was estimated using Spearman's ρ . To adjust for multiplicity in this preliminary analysis, associations with $P < 0.01$ were considered statistically significant. Several associations were found between proteomic markers and the first functional principal component for FEV₁ (lower values of this component correspond to milder disease); several markers including glutamate receptor 2 isoform 4, alpha-1-acid glycoprotein 2 precursor, immunoglobulin alpha-1 heavy chain constant region, anti-Entamoeba histolytica immunoglobulin kappa light chain, anti-HBsAg immunoglobulin Fab kappa chain, AT-rich interactive domain-containing protein 4A isoform I, and uridine-cytidine kinase-like 1 isoform 2 were positively associated (range of ρ : 0.37 to 0.69); additional markers including unconventional myosin-XVIIIa isoform X1, chromosome 12 OPR 42, isoform CRA c, and Keratin 18 were negatively associated (range of ρ : -0.73 to -0.54). Degree of rapid decline was negatively associated with markers A-kinase anchor protein 3 and Keratin 18 (ρ : -0.76 and -0.72, respectively). Patients who experienced rapid decline at a younger age tended to have elevated levels of markers alpha-1-acid glycoprotein 2 precursor, anti-Entamoeba histolytica immunoglobulin kappa light chain, and anti-HBsAg immunoglobulin Fab kappa chain, corresponding to negative associations (range of ρ : -0.55 to -0.43), and lower levels of chromosome 12 open reading frame 42, isoform CRA c (ρ : 0.55) (see FIG. 6).

The disclosed dynamic prediction model uses patient-specific information from the EPIC cohort while incorporating parameter estimates from Applicant's recent large-scale CFF-PR study to predict the onset of rapid decline. The model takes into account observed lung function for the patient at each time point, the mean FEV₁ evolution for each patient, and encompasses covariates with corresponding association parameters. Normal distribution of the data provides patient-specific heterogeneity between FEV₁ trajectories. Furthermore, a stochastic process is used to reflect the “saw tooth” variation over time for individual patients (see FIG. 1); integrated Brownian motion is used to depict this process. Normally distributed measurement error from the PFT is accounted for. Finally, data for the level of markers is added to the model. The covariance functions and estimation algorithm have been described previously (40). The model for the CFF-PR is implemented using the lme4 package (48) in R (R Foundation for Statistical Computing, Vienna, Austria). The covariates that considered from the CFF-PR may be expanded to include additional factors available from the EPIC study. Candidate molecular markers can be directly added as covariates, and the association parameters can be denoted to fit the model on smaller data. If marker data were available from all CFF-PR subjects, this new model could be called the “Full Model”. A so-called “Reduced Model” can be fit using detailed data from the EPIC cohort.

A novel semiparametric “big data” calibration approach (49) may be adapted to examine predictive value of the markers. The approach can be geometrically described (FIG. 8). The algorithm may be programmed using R. Briefly, constrained maximum likelihood estimation may be implemented to fit the Reduced Model described above while accounting for the parameter estimates found from the External Model on the CFF-PR. This estimation, as shown by Chatterjee, produces unbiased estimates with lower standard errors for the Full Model, compared to fitting the

Reduced Model alone. The calibrated estimates may be used to determine whether the addition of molecular markers and other EPIC covariates improve prediction of rapid decline. Improved prediction may be assessed by 1) examining the interaction of a given biomarker with age in the Reduced Model; 2) using model diagnostics; 3) fit statistics (e.g., AIC, BIC); 4) the likelihood ratio test to model; an example of the risk function for a patient is in FIG. 9. To validate predictive accuracy, analysis with 20% of the EPIC cohort removed from the modeling may be performed. This subset of the cohort will have covariate data added to the model to check predictive accuracy; metrics here may include mean absolute deviation (MAD) and mean square error (MSE), which have been previously used by Applicant with good success (35).

The predictive value of a subset of proteomic markers is assessed based on the Reduced Model using FD and dynamic prediction as described above. Applicant included the protein similar to dual specificity phosphatase 9, partial as a covariate and its interaction with age in the Reduced Model. Other covariates included in the model were age-specific components to fit the FEV₁ trajectory, as shown in the FD analysis in previous studies, and the mild/severe designation from the original study. The proteomic marker was available on 36 EPIC subjects (23 Mild, 13 Severe); these subjects were 6.3 (6.6-7.7) years of age with FEV₁ of 95.5 (61-138) % predicted. The median (range) number of PFTs (per subject was 54 (20-93); per-subject follow-up was 11.5 (9.4-12) years. The model showed the proteomic marker was significantly associated with mean FEV₁ (coefficient: -8.03, SE: 4.07, $z=-2.0$, $P=0.04$) and approached significance in its association with FEV₁ decline (coefficient for interaction: 1.03, SE: 0.79, $z=1.3$, $P=0.19$). These results suggest that the biomarker has a negative association with overall mean FEV₁ but may have a positive effect on rate of FEV₁ decline.

Covariate selection may also be performed using Bayesian Ensemble Trees (BET) (50). BET was developed by Applicant as an approach to perform variable selection for modeling FEV₁ decline. This approach utilizes the ensemble of classification and regression trees (CART). Each constituent tree is estimated with a subset of similar data, and can be used for covariate selection and imputation of missing data (51). The form of each marker covariate will be modified to examine potential lagged effects on FEV₁ decline. For select markers, associations with FEV₁ decline can be simultaneously examined using joint models for high-dimensional data (52). To investigate the impact of the collection of markers on rapid decline, PCA may be used to develop composite score(s); the score data may then be included in the model as covariate(s). Alternatively, BET may be used to create step functions representing rapid decline. The step functions can be subsequently smoothed to represent rate of FEV₁ decline and construct individual probabilities of rapid decline similar to the dynamic prediction model or risk scores for rapid decline.

Four Bayesian penalized regression approaches for the previously described Gaussian linear mixed effects model with non-stationary covariance to account for the complicated structure of longitudinal lung-function data while simultaneously estimating unknown parameters and selecting protein isoforms to improve the prediction model. These penalized regression models induce variable selection by shrinking the coefficient of irrelevant variables toward zero while simultaneously keeping the relevant variables in the

final model. Different types of shrinkage priors are considered and evaluated to induce variable selection in a fully Bayesian framework.

Model setup: The main form of the Bayesian penalized regression model implemented for the non-stationary Gaussian linear mixed effect model can be written as:

$$Y_{ij} = \alpha_0 + \alpha_1 t_{ij} + \sum_{k=1}^p \beta_k X_{ik} + U_i + W_i(t_{ij}) + Z_{ij},$$

where Y_{ij} is lung function measurement for subject i at time point t_{ij} , $i=1, \dots, N$; $j=1, \dots, n_i$; $k=1, \dots, p$. The β_k 's are the coefficients of the proteomic biomarkers that we want to shrink towards zero to induce simultaneous variable selection and X_{ik} is the covariate matrix for the proteomic biomarkers. Between-patient heterogeneity is incorporated in the model with a random intercept term U_i , where $U_i \sim N(0, \omega^2)$. The term $W_i(t_{ij})$ denotes realizations from the zero-mean, continuous-time integrated Brownian motion process, and $Z_{ij} \sim N(0, \tau^2)$ represents iid measurement error. This model can be written as, $Y_i \sim MVN(\psi, \alpha + X_i \beta, V_i(\phi))$, where $V_i(\phi) = \omega^2 J_i + \sigma^2 R_i + \tau^2 I_i$, and it is reparametrized as $V_i(\phi) = \tau^2 (\phi_1 J_i + \phi_2 R_i + I_i)$ where J_i is $n_i \times n_i$ matrix of ones, I_i is $n_i \times n_i$ identity matrix, and R_i is $n_i \times n_i$ matrix with $(j, k)^{th}$ element is,

$$\frac{\min(t_{ij}, t_{ik})^2}{2} \left(\max(t_{ij}, t_{ik}) - \frac{\min(t_{ij}, t_{ik})}{3} \right).$$

To obtain different types of penalized regression models, the following priors are assigned to complete model specification;

Bayesian Least Absolute Shrinkage Selection Operator (lasso): Preferable when the goal is obtaining a sparse set of potential predictive proteomic biomarkers. λ is the penalty parameter which controls the amount of penalty being applied to the coefficients of predictors (proteomic biomarkers) to force them near zero. The following prior distributions are assumed:

$$\beta | \tau^2, a_1^2, \dots, a_p^2 \sim MVN(0, \tau^2 D_L)$$

$$D_L = \text{diag}(a_1^2, \dots, a_p^2)$$

$$a_j^2 | \lambda \sim \frac{\lambda^2}{2} e^{-\lambda^2 a_j^2 / 2}$$

$$\lambda^2 \sim \text{Gamma}(r, \delta).$$

Bayesian Adaptive Lasso: Similar to the Bayesian Lasso described above but assigns different amount of penalty to different coefficients based on their importance.

$$a_j^2 | \lambda \sim \frac{\lambda^2}{2} e^{-\lambda^2 a_j^2 / 2}$$

$$\lambda_j^2 \sim \text{Gamma}(r, \delta).$$

Bayesian Ridge Regression: Bayesian ridge regression is more preferable when the predictors are highly correlated. When there exists multicollinearity between predictors, the

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ridge regression deals with collinearity and applies continuous shrinkage and improves prediction performance through a bias-variance trade-off.

$$\beta|\tau^2, \lambda \sim MVN(0, \tau^2 D_R)$$

$$D_R = \text{diag}(\lambda^2)$$

Bayesian Elastic-Net: This method merges the features of both the lasso and the ridge and takes into account both sparsity and correlation structure of the data. Hence, the elastic net simultaneously induces variable selection by forcing the coefficients of redundant variables toward zero and selection of groups of correlated covariates and handles with multicollinear predictors.

$$\beta|\tau^2, a_1^2, \dots, a_p^2 \sim MVN(0, \tau^2 D_{EN})$$

$$D_{EN} = \text{diag}\{(a_1^{-2} + \lambda_2)^{-1}, \dots, (a_p^{-2} + \lambda_2)^{-1}\}$$

$$a_j^2|\lambda_1 \sim \frac{\lambda_1^2}{2} e^{-\lambda_1^2 a_j^2/2}$$

$$\lambda_1^2 \sim \text{Gamma}(r_1, \delta_1) \text{ and } \lambda_2^2 \sim \text{Gamma}(r_2, \delta_2).$$

Further, assigned are non-informative uniform prior for α_0 and α_1 , flat uniform priors for ϕ_1 and ϕ_2 , non-informative scale invariant prior

$$\frac{1}{\tau^2}$$

for τ^2 under all four models mentioned above.

Given the priors and the data likelihood, the posterior distributions of all unknown parameters are obtained by using Bayes' theorem. Unknown parameters and hyperparameters of each model are estimated by sampling from their conditional posterior distributions through MCMC algorithm. Each MCMC algorithm is creating code in R software.

The need for analysis of samples collected from subject undergoing modulator therapy is critical, as CFTR modulation has become the standard of care, and this will likely influence the behavior of molecular markers.

Example 1. Improvement of Predicting Lung Function Decline

A number of interventional therapies are used to stave off lung function decline in patients with cystic fibrosis, including antibiotics, anti-inflammatory therapies, hypertonic saline to increase clearance, and others. The largest challenge in CF care is projecting lung function performance and intervening accordingly. Currently, intervention occurs when clear declines in lung function as measured by FEV₁ and other lagging indicators are observed. The approach of intervening once lung function has already declined rarely restores lost lung function; therefore, intervening before lung function decline occurs can be beneficial by preventing decline. An example that captures the range of the performance of some of the markers in Table 1 at improving the prediction of lung function is shown in FIG. 11.

Tandem mass spectrometry or ELISA may be used to measure the levels of Ig lambda and Ig kappa chains in collections of blood from a CF patient during their regular hospital visits. The levels of these proteins would be inserted

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into the algorithm described herein along with the measurements of FEV₁ for the patient over a number of hospital visits. The markers can be used individually or in combination. The algorithm will return a score that reflects the risk of lung function decline in the next 18 months. Based on the score returned a physician may choose to be more aggressive with conventional anti-inflammatory therapy to reduce or prevent impending lung function decline or follow the usual regiment of care if the score does not indicate a high risk of future lung function decline.

Example 2. Measurement of CFTR to Assess the Success of the Delivery of CFTR Gene Therapy

A large challenge for gene therapy with the CFTR gene is the lack of effective measures of the delivered CFTR transgene expression. The technology described herein allows us to measure the levels of full length or any portion of CFTR protein. FIG. 12 shows difference measured in the levels of CFTR between patients homozygous for the F508del mutation compared with the levels measured in the serum of patients heterozygous for F508del and various mutations that express minimal levels of protein. The difference observed by the technology claimed here of 1.65-fold reflects the expected higher levels of CFTR expression in the homozygous group. An example of using this technology would be in the context of a CFTR gene therapy trial. In such a trial, measuring the expression of the delivered transgene CFTR would be necessary. The technology described herein may be used to measure level of full length or a portion of CFTR in the serum, plasma, and/or bronchoalveolar lavage to assess the efficiency of gene delivery. An increase in levels of CFTR in serum, for example, would indicate successful delivery of CFTR.

Example 3

Rigorous individual monitoring can reduce treatment costs, improve clinical outcomes, and decrease disease progression rate in CF patients. The disclosed methods may be used to predict or identify individuals most likely to benefit from various interventions allowing for a proactive (vs. the current reactive) approach in managing pulmonary exacerbations for patients with CF.

Spirometry is the primary tool used to assess lung function in diseases, including asthma, chronic obstructive pulmonary disease, and CF. It has been central to defining the natural history of these disorders and in developing algorithms to predict their trajectories. (23) Spirometric measures (e.g., FEV₁) are the standard-of-care used to assess disease progression and the most common endpoints in clinical trials for lung disease. (25) However, they are insensitive to the cellular and molecular processes that drive lung disease.

While spirometry, primarily FEV₁, is the standard clinical measure of lung function and is used ubiquitously to monitor disease progression and therapy response in CF, measures are abnormally low in virtually all adult CF patients, but young patients usually display normal FEV₁. However, structural abnormalities, including bronchiectasis, are observed via computed tomography (CT) in up to 1/3 of CF patients before the 3rd year of life. (8). Similarly, bronchoalveolar lavage (BAL) has shown that infection and inflammation can be present years prior to clinical symptoms and spirometric changes. These observations indicate that common clinical metrics (e.g., body mass index and FEV₁) are trailing indicators of disease progression and represent the

culmination of years of pathological changes at the structural and molecular levels (10, 11). As such, they typically cannot be used to identify young patients at risk for rapid lung function decline or patients for whom interventions could prevent permanent lung disease from becoming established.

More sensitive clinical tools available to assess lung disease are also ill-suited for prospective monitoring in early CF. For example, BAL fluid is obtained via bronchoscopy, which is invasive and often requires sedation—particularly in pediatric subjects. As such, it cannot be used for intensive monitoring. Similarly, exposure to the ionizing radiation of CT is associated with increased risk of cancer in pediatric subjects, because children are more susceptible to radiation and have longer remaining lifespans than adults (12), making CT also ill-suited for frequent prospective monitoring in CF.

The duration of this spirometric “silent period” in CF lung disease is likely to increase in the coming years, further in view of the recently-approved triple-combination CFTR modulator therapy (Trikafta®, a combination of three drugs—elexacaftor/ivacaftor/tezacaftor—that target the defective CFTR protein). When treated with highly effective modulator therapy, CF patients display initially improved lung function and slower rates of decline, as assessed by FEV₁ (11). However, structural remodeling—in particular bronchiectasis—continues in these patients (13). Thus, the CF field is at a crossroads, where dramatic past improvements in care now limit the ability to assess early disease severity and predict disease progression. This is particularly true in young patients for whom effective interventions will yield the greatest improvements in outcomes and generate the largest increases in lifespan. As such, it is advantageous to provide sensitive methods to quantify disease progression and therapy response in the context of spirometrically normal lung function, which may further be non-invasive and radiation-free. Without such tools, further improvements in CF care and increases in CF patient longevity may be limited. The present disclosure seeks to address this gap with methods that may be used to monitor and predict lung function decline and structural remodeling in early CF.

The disclosed methods, in one aspect, may be used to fill this void by combining structural and functional lung measurements that may be used to detect, monitor, or predict lung pathophysiology. Exemplary methods useful for determining a structural or functional lung measurement include, but are not limited to, MRI, CT, or other imaging methods known in the art, which may be combined with FD analysis, and methods for protein biomarker measurement including, not limited to, LC-MS/MS (Liquid Chromatography incorporated with tandem Mass Spectrometry) and ELISA. In one exemplary embodiment, that reflect the mechanisms driving lung function decline and remodeling in CF. Previously, bronchiectasis markers (29-30) examined in a biased fashion, focused only on targeted proteins in blood using epitope-based analyses with aptamers, which are prone to false positives and negatives. These factors con-

tributed to only modest correlations ($r \sim 0.45$) and did not further the understanding of CF bronchiectasis from what was known from other lung diseases. In contrast, Applicant has identified markers that correlate with disease severity (mean $r \sim 0.72$) during stable disease, improving the ability to forecast lung function decline (FIG. 4).

LC-MS/MS discovery of systemic disease biomarkers and FD analysis: Based on cancer and cardiovascular research, where conventional “shotgun” proteomics identify 500-1000 proteins per sample but failed to distinguish disease severity cohorts, Applicant reasoned that deeper proteomic screening would be necessary for CF studies. Thus, Applicant developed a pipeline using multidimensional protein separation by abundant protein-adsorption columns and gel and column chromatography. This more rigorous approach generated multiple fractions from each sample for subsequent MS analysis (7-17). Cohorts of 44 mild (FEV₁>85%) and 44 severe (<45th percentile) CF patient were matched based on age, gender, genotype, and infection status, then randomized and blinded before proteomic analysis. Applicant identified 61,942 protein isoforms expressed in both cohorts, with 19,162 isoforms identified in all subjects. Principal Component Analysis (PCA), logistic regression with Least Absolute Shrinkage and Selection Operator (LASSO) (18), random forests (19), Wilcoxon Rank Sum (44), and McNemar’s tests (45) were used to reduce the dataset from ~62,000 isoforms analyzed to 744 isoforms across the battery of five tests. PCA of the top 744 isoforms with respect to FEV₁ indicated that these markers segregated mild from severe disease and are good discriminators of lung function in this cross-sectional group. A subset of isoforms were identified by nonbiased proteomic analysis and may be biomarkers of choice for developing a lung function decline prediction model. Building on these approaches, Applicant used FD analysis to characterize lung function at the population level and to accurately predict rapid decline in individual patients. For example, FIG. 14 shows the forecasting performance of the top 3, (voltage gated calcium channel alpha 1F subunit CACNA1F [9606], pleckstrin homology domain-containing family A member 5 isoform X1 PLEKHA5 [9606], and pleckstrin homology domain-containing family A member 5 isoform X9 PLEKHA5 [9606]).

In certain aspects, FEV₁ may be used with biostatistical modeling as a point-of-care tool (14). The predictive power may be improved when data from Liquid Chromatography is incorporated with tandem Mass Spectrometry (LC-MS/MS), which can yield proteomic biomarkers sensitive at the molecular level to changes in CF lung disease (FIG. 4, 5). Here, these biostatistical and proteomic tools may be combined with one or more biomarkers as disclosed herein, (See Table 4, Panel of protein biomarker covariates for algorithm). Functional information (e.g., as obtained via hyperpolarized (HP)¹²⁹Xe MRI, fluorine MRI, CT, or the like) may be used to determine regional lung dysfunction longitudinally and ultra-short echo-time (UTE), and may further be used to detect structural remodeling over time.

TABLE 4

Panel of protein biomarker covariates
Protein name (gene name)

1. alpha 1 type XXIV collagen precursor
2. alpha-2-macroglobulin
3. alpha-fetoprotein enhancer binding protein
4. Alstrom syndrome protein 1 isoform 1
5. aminopeptidase O isoform X9 (C9orf3)

TABLE 4-continued

Panel of protein biomarker covariates
Protein name (gene name)
6. ankyrin-3 isoform 3
7. anti-HBsAg immunoglobulin Fab kappa chain, partial
8. anti TNF-alpha antibody light-chain Fab fragment, partial
9. ARAP2 protein (ARAP2)
10. Bifunctional glutamate/proline—tRNA ligase (EPRS)
11. C10ORF6
12. CECR2 protein (CECR2)
13. cell division cycle 2-like 5 (cholinesterase-related cell division controller) (CDC2L5)
14. chromodomain-helicase-DNA-binding protein 6 isoform X3 (CHD6)
15. chromosome 3 open reading frame 15 (C3orf15)
16. cilia- and flagella-associated protein 91 isoform 3
17. cingulin-like 1 (CGNL1)
18. CLIP-associating protein 2 isoform X24 (CLASP2)
19. Coiled-coil domain-containing protein 18
20. coiled-coil domain-containing protein 18 isoform X2 (CCDC18)
21. coiled-coil domain-containing protein 93 isoform X3 (CCDC93)
22. collagen triple helix repeat-containing RP11-45B20.2
23. collagen, type XXIV, alpha 1 (COL24A1)
24. complement component C2 (C2)
25. complement component C6 isoform X7 (C6)
26. complement factor H (CFH)
27. Connector enhancer of kinase suppressor of ras
28. connector enhancer of kinase suppressor of ras 2 isoform X2 (CNKSR2)
29. CTP:phosphocholine cytidylyltransferase
30. death inducer-obliterators-3 (DIDO3)
31. diacylglycerol kinase iota isoform X4 (DGKI)
32. DNA annealing helicase and endonuclease ZRANB3 (ZRANB3)
33. DNA methyltransferase 1-associated protein 1
34. DNAH10 variant protein
35. E3 ubiquitin-protein ligase MIB2 isoform X5 (MIB2)
36. EPRSN1
37. fibrinogen alpha chain preproprotein, isoform alpha (FGA)
38. filamin 2 (FLN2)
39. Fras1-related extracellular matrix protein 2 (FREM2)
40. FYVE, RhoGEF and PH domain-containing protein 5 isoform X2 (FGD5)
41. Gelsolin
42. gelsolin isoform X3 (GSN)
43. glutamate receptor-interacting protein 1 isoform X7 (GRIP1)
44. growth-inhibiting protein 24 (GIG24)
45. hCG1656772
46. hemicentin
47. Histidine-rich glycoprotein
48. histidine-rich glycoprotein isoform X1 (HRG)
49. histone acetyltransferase MORF alpha
50. IF2 protein (IF2)
51. immunoglobulin lambda light chain VLJ region (IGL)
52. interferon regulatory factor-2 binding protein 2B
53. junction-mediating and -regulatory protein isoform X2 (JMY)
54. KIAA0328 protein KIAA0328
55. kinesin-like protein (KIF2)
56. kinesin-like protein KIF21A isoform X7 (KIF21A)
57. laminin subunit alpha-2 isoform X3 (LAMA2)
58. la-related protein 1 isoform X5 LARP1
59. LIM and calponin homology domains-containing protein 1 isoform X20 (LIMCH1)
60. LIM and calponin homology domains-containing protein 1 isoform X22 (LIMCH1)
61. lysine-specific demethylase 2B isoform X7 (KDM2B)
62. MAX gene-associated protein isoform X11 (MGA)
63. multiple endocrine neoplasia type 1 candidate protein number 18 (HSPF2)
64. MYCBP-associated protein isoform X1 (MYCBPAP)
65. nebulin-related anchoring protein, isoform CRA_b
66. Non-canonical poly(A) RNA polymerase PAPD5 (PAPD5)
67. PAP associated domain containing protein 5 variant
68. partitioning-defective 3-like protein splice variant c (PAR3L)
69. paraoxanase-3
70. PF6
71. Platelet glycoprotein 1b alpha chain (GP1BA)
72. Pleckstrin homology-like domain family B member 2 (PHLDB2)
73. PMFBP1 protein (PMFBP1)
74. poly (ADP-ribose) glycohydrolase (PARG)
75. PRO2841
76. Pregnancy zone protein
77. probable E3 ubiquitin-protein ligase HERC1 isoform X1 (HERC1)
78. probable E3 ubiquitin-protein ligase HERC1 isoform X11 (HERC1)
79. Protein AMBP (AMBP)
80. protein arginine N-methyltransferase 3 isoform X1 (PRMT3)
81. Protein SCAF11 (SCAF11)
82. ral GTPase-activating protein subunit alpha-1 isoform X12 (RALGAP1)

Panel of protein biomarker covariates	
Protein name (gene name)	
83.	Receptor-type tyrosine-protein phosphatase C (PTPRC)
84.	RGS3 isoform C2PA-RGS3 (RGS3)
85.	Rho-GTPase activating protein 10 (ARHGAP10)
86.	RNA-binding motif protein, Y chromosome, family 1 member B isoform X1 (RBM1B)
87.	RUNDC1 protein (RUNDC1)
88.	protein-methionine sulfoxide oxidase MICAL3 isoform X10 (MICAL3)
89.	Protein phosphatase 1 regulatory subunit (PPP1R12B)
90.	Protein sidekick-2 (SDK2)
91.	sacsin isoform X3 (SACS)
92.	SAPS domain family, member 2 (SAPS2)
93.	serine/threonine-protein phosphatase 4 regulatory subunit 3A isoform 1
94.	serine-protein kinase ATM isoform X5 (ATM)
95.	serum arylalkylphosphatase precursor
96.	SHMT2 protein (SHMT2)
97.	smoothelin-B3 (SMTN)
98.	somatostatin receptor interacting protein splice variant a (SSTRIP)
99.	sperm-associated antigen 17 isoform X3 (SPAG17)
100.	spermatogenesis-associated protein 31C1 isoform X1 (SPATA31C1)
101.	synaptotagmin-like protein 2 isoform X10 (SYTL2)
102.	T-cell lymphoma invasion and metastasis 1 (TIAM1)
103.	translation initiation factor (IF2)
104.	transferrin precursor
105.	terminal nucleotidyltransferase 4B isoform b
106.	tetratricopeptide repeat protein 21A isoform X6 (TTC21A)
107.	thrombospondin 1 (THBS1)
108.	triadin isoform X17 (TRDN)
109.	ubiquitin carboxyl-terminal hydrolase 40 isoform X1 (USP40)
110.	uridine-cytidine kinase-like 1 isoform 2
111.	uridine-cytidine kinase-like 1 isoform X7 (UCKL1)
112.	utrophin isoform X4 (UTRN)
113.	Voltage-dependent N-type calcium channel subunit alpha-1B (CACNA1B)
114.	WD repeat-containing protein 64 isoform X2 (WDR64)
115.	zinc finger and AT hook domain containing (ZFAT)
116.	Zinc finger homeobox protein 4 (ZFHX4)

For predictive modeling, the primary outcome—lung function—may be defined by the ventilation defect percent (VDP) or other regional measures (partial ventilation, ventilation heterogeneity, hyper-intensity, etc.). The FD model (46) may be used to analyze longitudinal VDP as a function of time (indexed as age at the time of functional assessment, via MRI scan, CT, or the like, and blood draw) and degree of proteomic expression with adjustment for select clinical/demographic characteristics as covariates, including lung clearance index (LCI). Applicant has built a prediction model using patient-specific information to predict the onset of rapid decline. For VDP, this model can be expressed as Equation 9:

$$VDP_{ij} = \mu_i + W_i(t_{ij}) + Z_i(t_{ij}) = f(t_{ij}) + X_{ik}\theta_k, \quad (\text{Equation 9})$$

where VDP_{ij} is the ventilation metric for the i^{th} patient at time point t_{ij} (age in years). The function $\mu_i(t_{ij})$ is mean VDP evolution for the i^{th} patient, which includes spline formulation $f(t_{ij})$ to characterize overall VDP progression as FD and encompasses covariates $X_{i1}, \dots, X_{ip}$, with corresponding association parameters $\theta_{i1}, \dots, \theta_{ip}$. The term U_i is assumed to follow a normal distribution with mean 0 and variance σ^2 , and provides patient-specific variability. $W_i(t_{ij})$ is a stochastic process reflecting image variation over time in an individual patient. Integrated Brownian motion is used to depict this process, which has variance σ^2 . Z_i represents normally distributed measurement error with mean 0 and variance τ^2 . The covariance functions and estimation algorithm have been described previously (47). Predictive models for other regional lung function metrics can be constructed similarly. The model for the CFF-PR was implemented using the Imenssp package (48) in R (R Foundation for Statistical Computing, Vienna, Austria). Candidate molecular markers

$B_{i1}, \dots, B_{im}$ may be used as covariates (main effect and interaction effect with time), and denote the association parameters as $\gamma_1, \dots, \gamma_m$ to fit the model on smaller data.

When using structural remodeling as a covariable, which may include data obtained via any imaging method known in the art, including, but not limited to, CT, MRI, or other image analysis technique, prediction modeling and evaluation may be used, except the discrete response variable will be log-transformed and expressed as Equation 10:

$$\log(\text{UTE}_{\text{bronch}}) = \mu_i(t_{ij}) + U_i + W_i(t_{ij}) + Z_i, \quad (\text{Equation 10})$$

One outcome, for example, the degree of bronchiectasis, can be defined by expert reader scoring or using automated software (e.g., deep learning algorithms), may use scores from FPCA to estimate correlation between proteomic expression and degree of bronchiectasis (49).

Sensitivity of MRI/CT to early CF lung disease: The disclosed modeling may be applied to advanced lung imaging data, which can be performed in children too young to perform spirometry (50) and detect pathology prior to spirometric changes (FIG. 18). Any acquisition strategy can be used. For example, in one approach an efficient 3D spiral, FLORET (51-55) sequence to depict ventilation at 3-mm resolution may be used the highest ^{129}Xe resolution achieved in humans. “Keyhole” reconstruction pipeline may be used to mitigate regional ^{129}Xe signal decay and improve image accuracy (56). Biomarkers of structural remodeling using a ^1H FLORET acquisition may be optimized to visualize airway abnormalities. This free-breathing protocol generates images with 1-mm resolution in ~5 minutes (57, 58). Airway-optimized ^1H FLORET may be used to guide clinical bronchoscopies to obtain bronchoalveolar lavage (BAL) fluid from specific lung pathologies, including bron-

chiectasis. Using these data and FD analysis, molecular changes may be used to forecast functional and structural disease in individual CF patients. These methods may combine innovations in proteomics, imaging and modeling to 1) enable translational research that continues to advance CF patient care and 2) define the post-CFTR-modulator “new normal” for the natural history of CF. While the markers identified in this work may be specific to CF, combining non-invasive HP ^{129}Xe and UTE MRI with blood and BAL proteomics may be applicable to a wide range of lung diseases.

Functional Data (FD) and Functional Principal Component (FPC) analyses: The disclosed methods build on FD analysis and longitudinal models using FEV₁ data from the CFF-PR. This modeling blends established biostatistical approaches with modern FD analyses to characterize the nonlinear FEV₁ trajectory of individual patients and predicts subsequent decline. The predictive models may be used for in-clinic applications for decision aids for treatment planning and at the time of patient encounters. Furthermore, predictive accuracy of lung function decline improves with proteomic markers (FIG. 14). Biomarker data correlation with FEV₁ may be measured in multiple simulations and by appropriate statistical tests as described.

Hyperpolarized (HP) ^{129}Xe MRI: Images from 50 CF patients (24 male, 26 female) were acquired. HP ^{129}Xe polarization (~20-40%) was measured before MRI with a polarimetry station. Images were acquired with ^{129}Xe coils to provide comparable performance across the range of subject sizes (15). Single-breath HP ^{129}Xe images were acquired using a multi-slice gradient-echo sequence, linear phase encoding, and optimized flip angle (16). Impaired ventilation was assessed using the Ventilation Defect Percentage (VPD), defined by employing automated software to identify lung voxels with signal intensities below a threshold that maximized contrast between age-matched, healthy control subjects and CF patients (17).

UTE MRI: Historically, lung MRI has been challenging, because magnetic inhomogeneities are created by air-tissue interfaces (i.e., alveoli) and cause the MR signal to decay with a rate constant (T_2^*) of ~0.8 ms at 3 T (18). Rapid decay was mitigated using an ultra-short echo-time (UTE) sequence to sample the MR signal before significant decay (echo-times of <200 μs), retaining ~80% of parenchymal signal. Data were collected at end expiration, where T_2^* and motion are minimized (19). Structural pathology (wall thickening, bronchiectasis, etc.) was identified and scored by two board-certified, pediatric radiologists using an MRI-variant of the well-established “Brody” scoring system (20) (FIG. 15).

Correlation of MRI and Proteomics: ^{129}Xe MRI and same-day blood draws were performed in 9 CF patients with mild lung disease (FEV₁>85%). Blood samples were grouped by functional impairment according to ventilation defect percentage (VDP) as no impairment (VPD <2, 2 male, 1 female), mild (5<VDP<15; 1 m, 2 f), and moderate (VDP >20, 2 m, 1 f). UTE MRI and same-day blood draws were performed in 8 CF patients with mild disease. Bronchiectasis was scored with a simplified system by dividing lungs into 6 regions (5 lobes plus lingula) (20, 21). Regions were scored independently by a trained reader (scale: 0-3; 0=not present; 1=present in <1/3 lobe; 2=present in <2/3 lobe; and 3=present in >2/3 lobe), and regional scores were summed to yield whole-lung scores. Samples were grouped by bronchiectasis severity as mild (score<3; 2 m, 2 f) or

moderate (4 score <10; 2 m, 2 f). For both UTE and VDP, group identities were blinded for non-biased proteomic analysis.

Correlation of MRI and Proteomics: For functional MRI studies, ^{129}Xe MRI and same-day blood draws were performed in 9 CF patients with mild lung disease (FEV₁>85%). Blood samples were grouped by functional impairment as no impairment (VPD<2, 2 male, 1 female), mild (5<VDP<15; 1 m, 2 f), and moderate (VDP>20, 2 m, 1 f). UTE MRI and same-day blood draws were performed in 8 CF patients with mild disease. For structural studies, bronchiectasis was scored with a simplified system by dividing lungs into 6 regions (5 lobes plus lingula) (21, 22). Regions were scored independently by a trained reader (scale: 0-3; 0=not present; 1=present in <1/3 lobe; 2=present in <2/3 lobe; and 3=present in >2/3 lobe), and regional scores were summed to yield whole-lung scores. Samples were grouped by bronchiectasis severity as mild (score<3; 2 m, 2 f) or moderate (4 score 10; 2 m, 2 f). For both UTE and VDP, group identities were blinded for non-biased proteomic analysis.

Using pathway, network, and process analyses, Applicant successfully identified the no, mild, or moderate impairment groups before data were unblinded. Analysis of proteomic changes revealed differences in pro-inflammatory signaling, remodeling-associated cytoskeletal rearrangement, and cell adhesion associated with inflammatory cell activation (FIG. 16). Analysis of the groups segregated by structural Ultra-Short Echo Time (UTE) MRI revealed changes in pathways associated with CF disease (e.g. ENAC regulation), inflammatory signaling, remodeling, and cell adhesion (FIG. 17). The no, mild, or moderate impairment groups were successfully identified by Applicant before data were unblinded. Of particular note—all subjects had 85% or higher FEV₁ percent predicted. Current pulmonary function tests cannot distinguish impairment in patients with 85% or higher FEV₁ percent predicted. Analysis of proteomic changes revealed differences in pro-inflammatory signaling, remodeling-associated cytoskeletal rearrangement, and cell adhesion associated with inflammatory cell activation (FIG. 16). Analysis of the groups segregated by structural Ultra-Short Echo Time (UTE) MRI revealed changes in pathways associated with CF disease (e.g. epithelial sodium channel (ENAC) regulation), inflammatory signaling, remodeling, and cell adhesion (FIG. 17). These blinded studies provide compelling evidence that proteomics detects mild structural and ventilation abnormalities identified by MRI, supporting Applicant’s hypothesis that these technologies yield robust and complimentary information to detect early remodeling and lung function impairment. Significantly, this sensitivity to early disease cannot be validated with available lung function testing.

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All percentages and ratios are calculated by weight unless otherwise indicated.

All percentages and ratios are calculated based on the total composition unless otherwise indicated.

It should be understood that every maximum numerical limitation given throughout this specification includes every lower numerical limitation, as if such lower numerical limitations were expressly written herein. Every minimum numerical limitation given throughout this specification will include every higher numerical limitation, as if such higher numerical limitations were expressly written herein. Every numerical range given throughout this specification will include every narrower numerical range that falls within such broader numerical range, as if such narrower numerical ranges were all expressly written herein.

The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as "20 mm" is intended to mean "about 20 mm."

Every document cited herein, including any cross referenced or related patent or application, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

The invention claimed is:

1. A method for treating an individual at risk for non-linear lung function decline, comprising

a) determining, in said individual, a covariate comprising one or more biomarker selected from Immunoglobulin alpha-1 heavy chain constant region; kappa 1 immunoglobulin constant; Immunoglobulin kappa light chain VLJ region; immunoglobulin lambda; Immunoglobulin lambda constant 2; Immunoglobulin lambda constant 3; Tight junction protein 3; Alpha-1-acid glycoprotein 2 precursor; Signal-induced proliferation-associated 1-like protein 3; Plakophilin 4; Inter-alpha (globulin) inhibitor H2; Absent in melanoma 1 protein; Actin filament-associated protein 1 isoform X2; Serine/threonine-protein phosphatase PP1-beta catalytic subunit; Retinol-binding protein 4; Fermitin family homolog 1; Actin, cytoplasmic 2; Transthyretin; Melanoma inhibitory activity protein 3 isoform X2; Pleckstrin homology domain-containing family G member 1 isoform X3; Pleckstrin homology domain-containing family A member 5 isoforms X1-X9; Pleckstrin; Zinc finger protein 295; Protocadherin Fat 2; Cadherin-related family member 2 isoform X1; Voltage gated

calcium channel alpha 1F subunit; Actin filament associated protein; Heat shock 70 kDa protein 1-like; EF-hand calcium binding protein 2; Polyamine modulated factor-1 (PMF1) protein; Keratin 16; Keratin 18; DNA replication ATP-dependent helicase/nuclease DNA2 isoform X4; Non-lens beta gamma-crystallin like protein; Bromodomain and WD repeat-containing protein 1 isoform X1; Collagen type V alpha 3 chain; Collagen type IV alpha; Dynein heavy chain 12, axonemal isoform X5; Serine/arginine repetitive matrix protein 1; Ras-related protein Rap-1A; AT-hook-containing transcription factor isoform 1; Coiled-coil domain-containing protein 18 isoforms X1-X7; Coiled-coil domain-containing protein 180; Patatin-like phospholipase domain-containing protein 2 isoform X1; Alstrom syndrome protein 1 isoform 1; aminopeptidase O isoform X9 (C9orf3); ankyrin-3 isoform 3; ARAP2 protein (ARAP2); Bifunctional glutamate/proline-tRNA ligase (EPRS); C10ORF6; CECR2 protein (CECR2); cingulin-like 1 (CGNL1); coiled-coil domain-containing protein 93 isoform X3 (CCDC93); complement component C2 (C2); complement component C6 isoform X7 (C6); complement factor H (CFH); connector enhancer of kinase suppressor of ras 2 isoform X2 (CNKSR2); CTP:phosphocholine cytidyltransferase; DNA annealing helicase and endonuclease ZRANB3 (ZRANB3); EPRSN1; fibrinogen alpha chain preproprotein, isoform alpha (FGA); filamin 2 (FLN2); Fras1-related extracellular matrix protein 2 (FREM2); FYVE, RhoGEF and PH domain-containing protein 5 isoform X2 (FGD5); gelsolin isoform X3 (GSN); glutamate receptor-interacting protein 1 isoform X7 (GRIP1); hCG1656772; histidine-rich glycoprotein isoform X1 (HRG); interferon regulatory factor-2 binding protein 2B; junction-mediating and -regulatory protein isoform X2 (JMY); KIAA0328 protein KIAA0328; 1a-related protein 1 isoform X5 LARP1; MAX gene-associated protein isoform X11 (MGA); paraoxanase-3; Platelet glycoprotein Ib alpha chain (GP1BA); PMFBP1 protein (PMFBP1); PRO2841; protein arginine N-methyltransferase 3 isoform X1 (PRMT3); Protein SCAF11 (SCAF11); ral GTPase-activating protein subunit alpha-1 isoform X12 (RALGAP1); Receptor-type tyrosine-protein phosphatase C (PTPRC); RGS3 isoform C2PA-RGS3 (RGS3); Rho-GTPase activating protein 10 (ARHGAP10); RNA-binding motif protein, Y chromosome, family 1 member B isoform X1 (RBM1B); protein-methionine sulfoxide oxidase MICAL3 isoform X10 (MICAL3); Protein sidekick-2 (SDK2); sacsin isoform X3 (SACS); SAPS domain family, member 2 (SAPS2); serum arylalkylphosphatase precursor; spermatogenesis-associated protein 31C1 isoform X1 (SPATA31C1); synaptotagmin-like protein 2 isoform X10 (SYTL2); translation initiation factor (IF2); triadin isoform X17 (TRDN); uridine-cytidine kinase-like 1 isoform X7 (UCKL1); utrophin isoform X4 (UTRN); WD repeat-containing protein 64 isoform X2 (WDR64); zinc finger and AT hook domain containing (ZFAT); Alstrom syndrome protein 1 isoform 1; aminopeptidase O isoform X9 (C9orf3); ankyrin-3 isoform 3; ARAP2 protein (ARAP2); Bifunctional glutamate/proline-tRNA ligase (EPRS); C10ORF6; CECR2 protein (CECR2); cingulin-like 1 (CGNL1); coiled-coil domain-containing protein 93 isoform X3 (CCDC93); complement component C2 (C2); complement component C6 isoform X7 (C6); complement factor H (CFH);

connector enhancer of kinase suppressor of ras 2 isoform X2 (CNKSR2); CTP:phosphocholine cytidyltransferase; DNA annealing helicase and endonuclease ZRANB3 (ZRANB3); EPRSN1; fibrinogen alpha chain preproprotein, isoform alpha (FGA); filamin 2 (FLN2); Fras1-related extracellular matrix protein 2 (FREM2); FYVE, RhoGEF and PH domain-containing protein 5 isoform X2 (FGD5); gelsolin isoform X3 (GSN); glutamate receptor-interacting protein 1 isoform X7 (GRIP1); hCG1656772; histidine-rich glycoprotein isoform X1 (HRG); interferon regulatory factor-2 binding protein 2B; junction-mediating and -regulatory protein isoform X2 (JMY); KIAA0328 protein KIAA0328; 1a-related protein 1 isoform X5 LARP1; MAX gene-associated protein isoform X11 (MGA); paraoxanase-3; Platelet glycoprotein Ib alpha chain (GP1BA); PMFBP1 protein (PMFBP1); PRO2841; protein arginine N-methyltransferase 3 isoform X1 (PRMT3); Protein SCAF11 (SCAF11); ral GTPase-activating protein subunit alpha-1 isoform X12 (RALGAP1); Receptor-type tyrosine-protein phosphatase C (PTPRC); RGS3 isoform C2PA-RGS3 (RGS3); Rho-GTPase activating protein 10 (ARHGAP10); RNA-binding motif protein, Y chromosome, family 1 member B isoform X1 (RBM1B); protein-methionine sulfoxide oxidase MICAL3 isoform X10 (MICAL3); Protein sidekick-2 (SDK2); sacsin isoform X3 (SACS); SAPS domain family, member 2 (SAPS2); serum arylalkylphosphatase precursor; spermatogenesis-associated protein 31C1 isoform X1 (SPATA31C1); synaptotagmin-like protein 2 isoform X10 (SYTL2); translation initiation factor (IF2); triadin isoform X17 (TRDN); uridine-cytidine kinase-like 1 isoform X7 (UCKL1); utrophin isoform X4 (UTRN); WD repeat-containing protein 64 isoform X2 (WDR64); zinc finger and AT hook domain containing (ZFAT);

- b) calculating a risk probability score based on said determining of one or more covariate, said risk probability score being used to characterize the individual as having no predicted lung impairment, mild predicted lung impairment, moderate predicted lung impairment, or severe predicted lung impairment wherein said risk probability score comprises a risk probability of said non-linear lung function decline in one or more parameters selected from a decrease in forced expiratory volume in one second (FEV1), pulmonary exacerbation (PE) frequency, FEV1-indicated exacerbation signal (FIES), ventilation defect percent (VDP), abnormal lung heterogeneity in lung as measured via imaging, lung hyperinflation, and combinations thereof; and
- c) treating said individual characterized as having mild predicted lung impairment, moderate predicted lung impairment, or severe predicted lung impairment with anti-inflammatory therapy.

2. The method of claim 1, said determining comprising determining an expression level of a protein of said biomarker.

3. The method of claim 1, further comprising determining an imaging marker covariate.

4. The method of claim 1, said imaging marker being a functional lung measurement, a structural lung measurement, or combinations thereof.

5. The method of claim 3, said imaging marker being obtained by an imaging method.

6. The method of claim 4, said imaging marker being detected using an image technique selected from hyperpo-

larized (HP) 129Xe, Ultra-short Echo-time (UTE) Magnetic resonance imaging (MRI), computed tomography (CT), and combinations thereof.

7. The method of claim 3, said imaging marker being structural remodeling.

8. The method of claim 3, said imaging marker being bronchiectasis.

9. The method of claim 1, said predicted lung impairment being a non-linear decline in one or more parameters selected from Ventilation Defect Percentage (VDP), FEV1, partial ventilation, ventilation heterogeneity, ventilation hyper-intensity or a combination thereof.

10. The method of claim 1, said risk probability score predicting the probability of lung function decline over a period selected from three weeks to four months, or six months, or twelve months.

11. The method of claim 1, wherein non-linear lung function decline is defined as a rate of change in longitudinal FEV1 that falls below 1.5% predicted/year.

12. The method of claim 1, wherein non-linear lung function decline is defined by the FEV1-indicated exacerbation signal (FIES) score.

13. The method of claim 1, further comprising detecting a one-or more covariates comprising a time-varying covariate.

14. The method of claim 13, said time-varying covariate selected from one or more of an infection with *P. aeruginosa*, an infection with Methicillin-resistant *Staphylococcus aureus* (MRSA), and Cystic-Fibrosis-related diabetes.

15. The method of claim 1, wherein said biomarker is one or more biomarkers selected from selected from Immunoglobulin alpha-1 heavy chain constant region; kappa 1 immunoglobulin constant; Immunoglobulin kappa light chain VLJ region; immunoglobulin lambda; Immunoglobulin lambda constant 2; Immunoglobulin lambda constant 3; Tight junction protein 3; Alpha-1-acid glycoprotein 2 precursor; Signal-induced proliferation-associated 1-like protein 3; Plakophilin 4; Inter-alpha (globulin) inhibitor H2; Absent in melanoma 1 protein; Actin filament-associated protein 1 isoform X2; Serine/threonine-protein phosphatase PP1-beta catalytic subunit; Retinol-binding protein 4; Fermitin family homolog 1; Actin, cytoplasmic 2; Transthyretin; Melanoma inhibitory activity protein 3 isoform X2; Pleckstrin homology domain-containing family G member 1 isoform X3; Pleckstrin homology domain-containing family A member 5 isoforms X1-X9; Pleckstrin; Zinc finger protein

295; Protocadherin Fat 2; Cadherin-related family member 2 isoform X1; Voltage gated calcium channel alpha 1F subunit; Actin filament associated protein; Heat shock 70 kDa protein 1-like; E F-hand calcium binding protein 2; Polyamine modulated factor-1 (PMF1) protein; Keratin 16 (KKRT16A); Keratin 18 (KRT18); DNA replication ATP-dependent helicase/nuclease DNA2 isoform X4; Non-lens beta gamma-crystallin like protein; Bromodomain and WD repeat-containing protein 1 isoform X1; Collagen type V alpha 3 chain; Collagen type IV alpha; Dynein heavy chain 12, axonemal isoform X5; Serine/arginine repetitive matrix protein 1; Ras-related protein Rap-1A; AT-hook-containing transcription factor isoform 1; Coiled-coil domain-containing protein 18 isoforms X1-X7; Coiled-coil domain-containing protein 180; and Patatin-like phospholipase domain-containing protein 2 isoform X1.

16. The method of claim 1, wherein said individual is diagnosed with cystic fibrosis (CF).

17. The method of claim 1 wherein said individual has a normal forced expiratory volume in one second (FEV1) as measured by spirometry.

18. The method of claim 1, wherein said individual is a pediatric patient.

19. The method of claim 1, wherein said individual is a patient under the age of 13 years of age.

20. The method of claim 1, wherein said method forecasts lung function trajectory for a period of about three weeks to about six months, or about six months to about twelve months.

21. The method of claim 1, further comprising applying a statistical algorithm to estimate correlation between said covariate value and predicted lung function.

22. The method of claim 1, wherein said method is carried out via a computer system, and wherein said method comprises capturing and displaying information related to said characterization of said individual using a graphical user interface (GUI).

23. The method of claim 1, further comprising assessing a variable selected from one or more of sex, body mass index (BMI), pulmonary exacerbation (PE), number of hospitalizations, antibiotic status, infection status, and combinations thereof, in said individual.

24. The method of claim 1, wherein said lung function decline is defined by one or more of absolute change in FEV1, rate of decline, risk of non-linear decline, and FIES.

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